Assignment-03

Course Code: CSE-1102 | **Course Title:** Introduction to Computing Sessional
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Batch: 68 | **Section:** E

PART C
Section 1: Introduction to Loops (Basic for Loop)

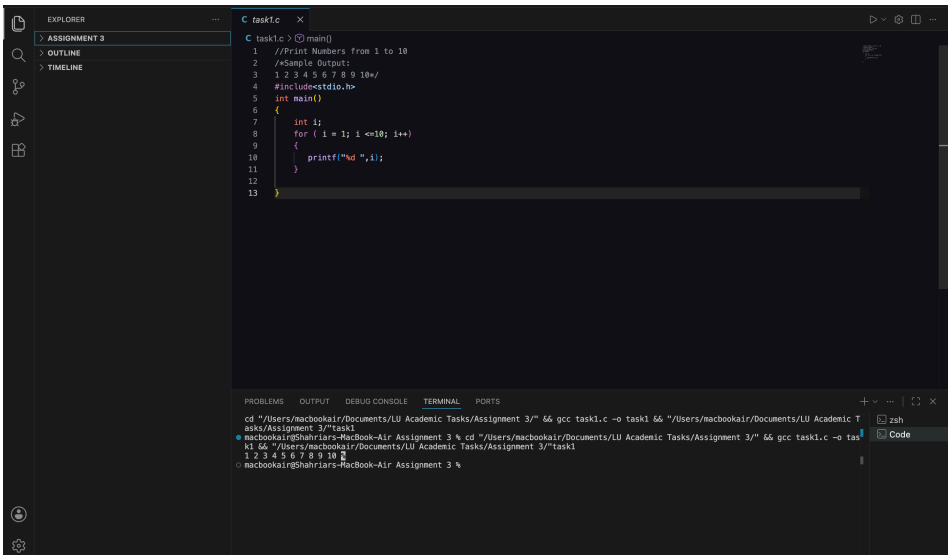
1. Write a C program using `for` loop to print numbers from 1 to 10.

Sample Output:
```
1 2 3 4 5 6 7 8 9 10
```

Program:

```c
#include<stdio.h>
int main()
{
    int i;
    for ( i = 1; i <=10; i++)
    {
       printf("%d ",i);
    }
     }
```

Output Screenshot:



2. Write a program using loop to print numbers from 1 to 10 in reverse order.

Sample Output:

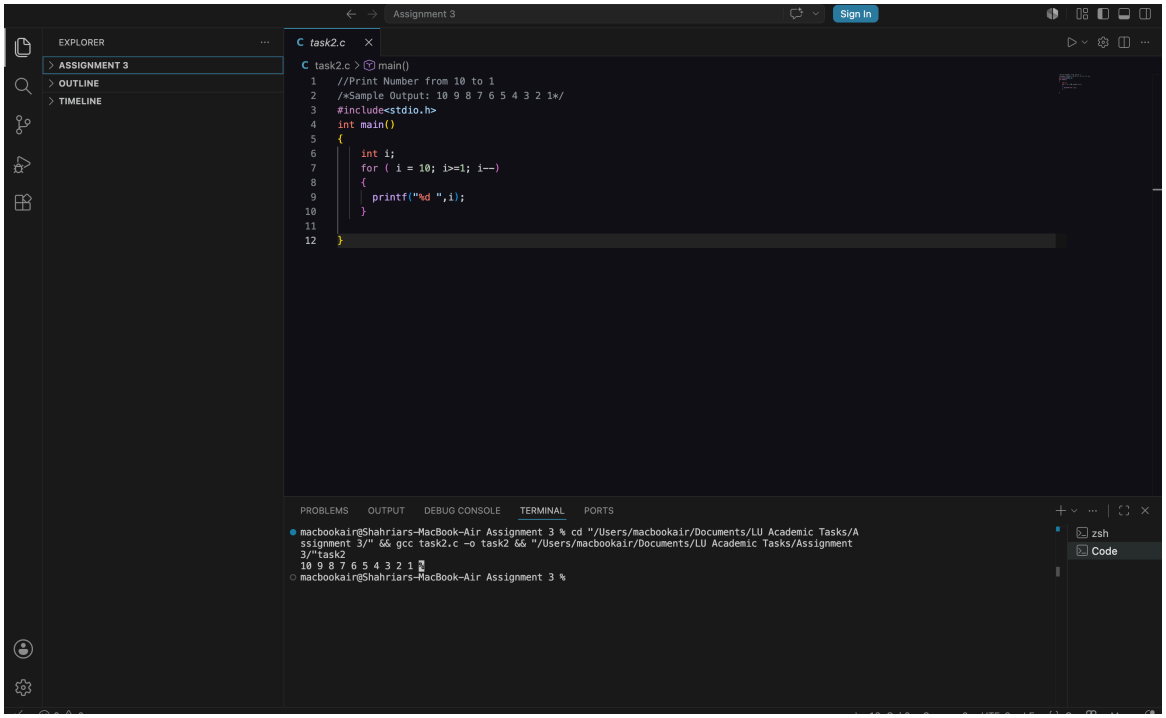
```
10 9 8 7 6 5 4 3 2 1
```

Program:

```
#include<stdio.h>
int main()
{
    int i;
    for ( i = 10; i>=1; i--)
    {
      printf("%d ",i);
    }

}
```

Output Screenshot:



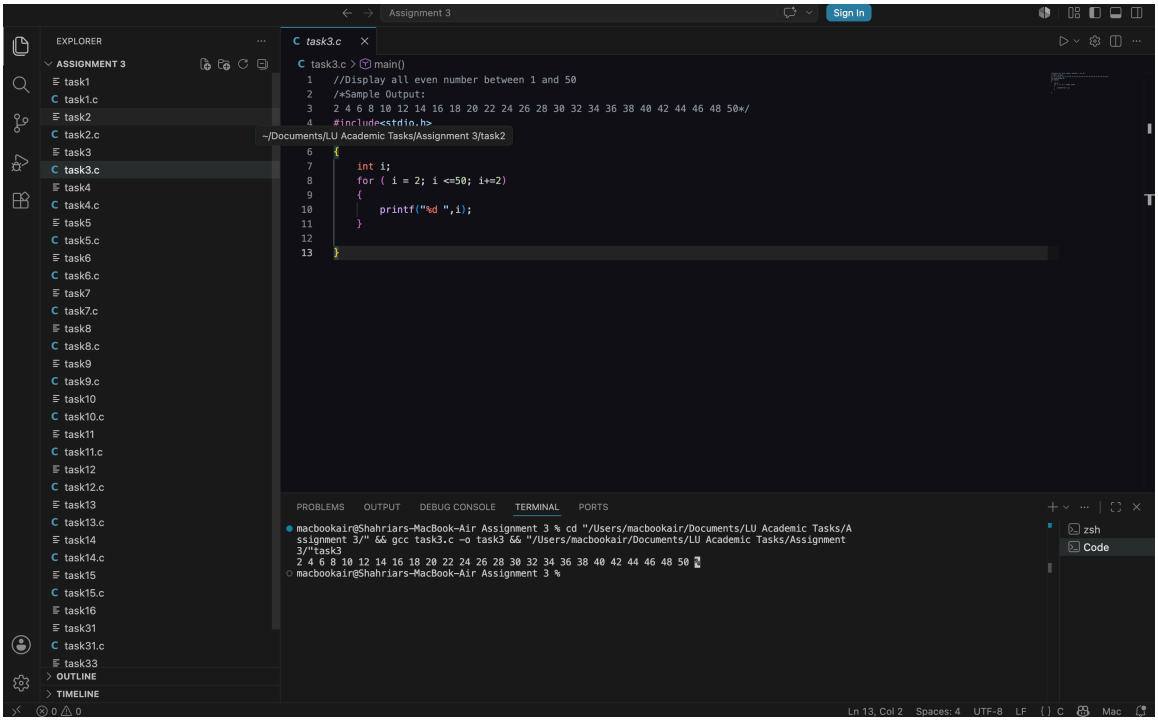
3. Display all even numbers between 1 and 50.

Sample Output:
```
2 4 6 8 10 ...
```

Program:

```c
#include<stdio.h>
int main()
{
    int i;
    for ( i = 2; i <=50; i+=2)
    {
        printf("%d ",i);
    }
}
```

Output Screenshot:



4. Display all odd numbers between 1 and 50.

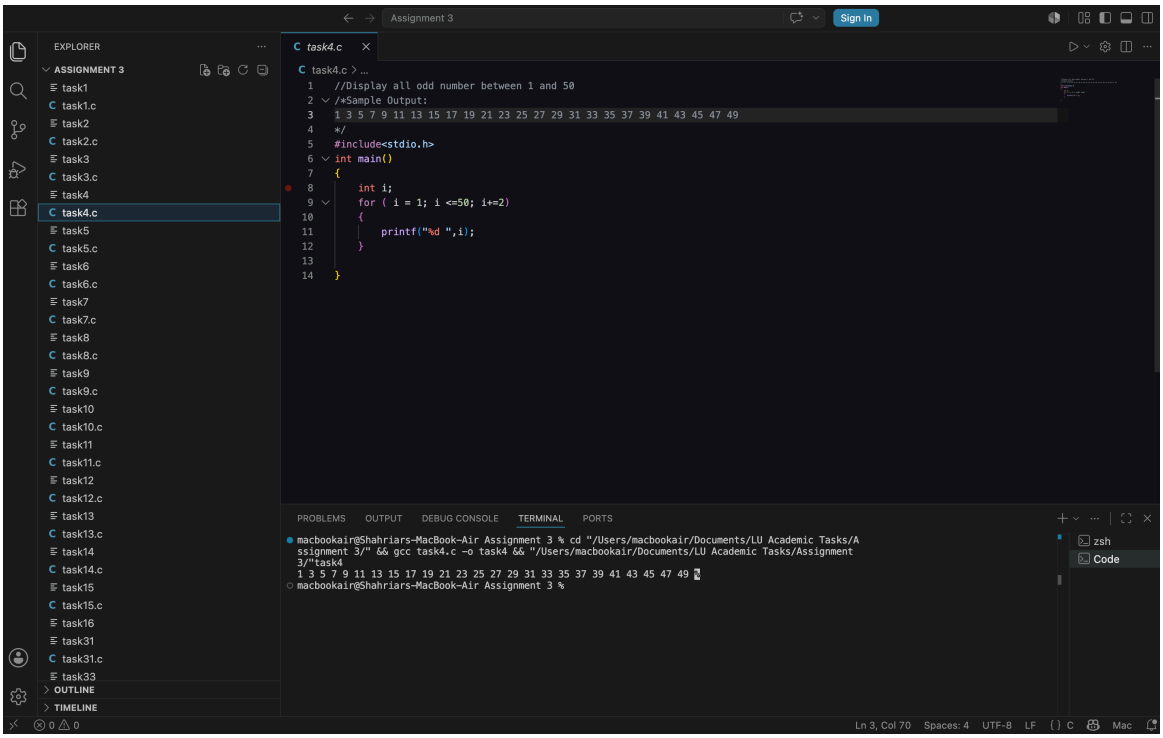
Sample Output:
```
1 3 5 7 9 …
```

Program:

```c
#include<stdio.h>
int main()
{
    int i;
    for ( i = 1; i <=50; i+=2)
    {
        printf("%d ",i);
    }

}
```

Output Screenshot:



5. Take a number `N` as input and calculate:

```
1 + 2 + 3 + ... + N
```

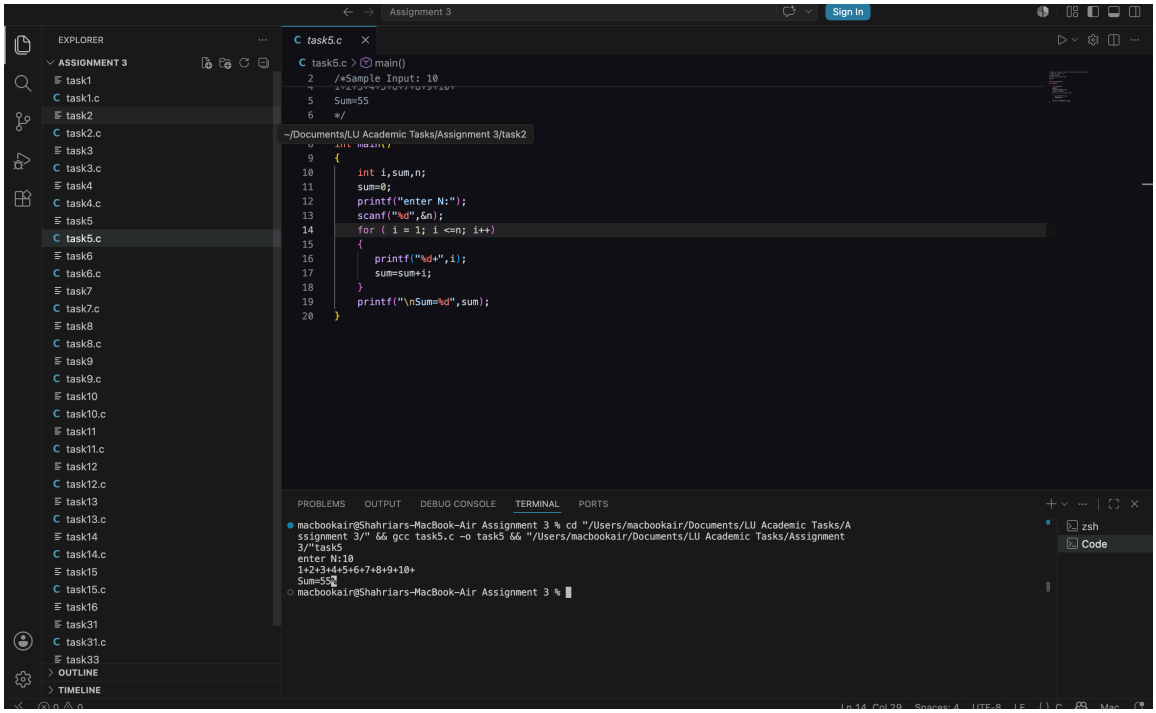
Sample Input:
```
10
```

Sample Output:
```
Sum = 55
```

Program:

```c
#include<stdio.h>
int main()
{
    int i,sum,n;
    sum=0;
    printf("enter N:");
    scanf("%d",&n);
    for ( i = 1; i <=n; i++)
    {
       printf("%d+",i);
       sum=sum+i;
    }
    printf("\nSum=%d",sum);
}
```

Output Screenshot:



Section 2: While Loop Applications

6. Take an integer and print its multiplication table using `while` loop.

Sample Input: 5

Sample Output:

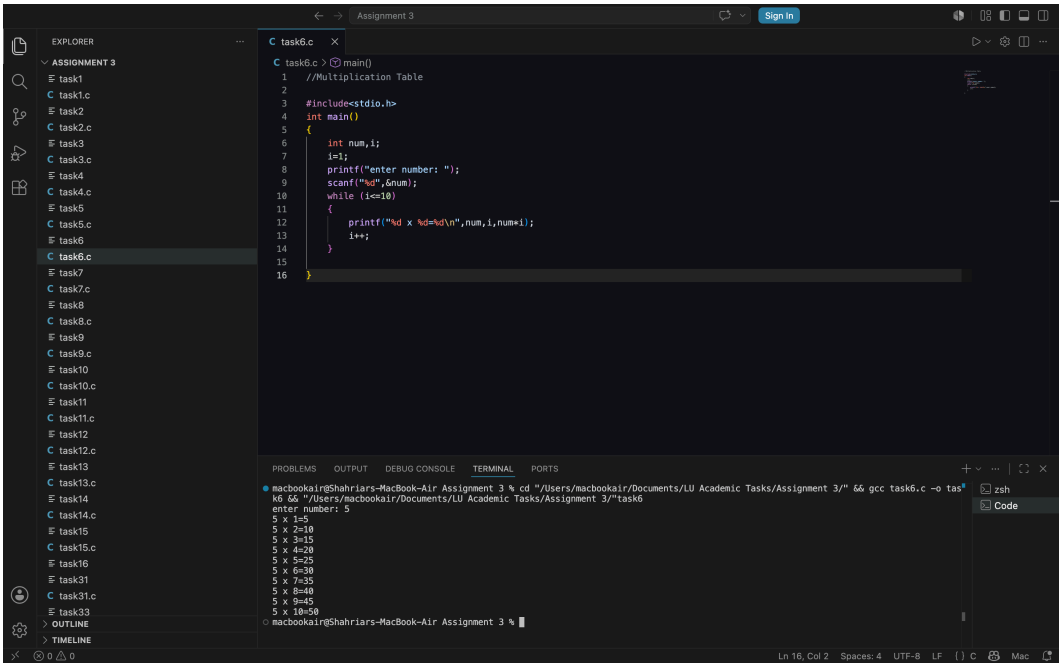
```
5 x 1 = 5
5 x 2 = 10
...
5 x 10 = 50
```

Program:

```c
#include<stdio.h>
int main()
{
    int num,i;
    i=1;
    printf("enter number: ");
    scanf("%d",&num);
    while (i<=10)
    {
        printf("%d x %d=%d\n",num,i,num*i);
        i++;
    }

}
```

Output Screenshot:

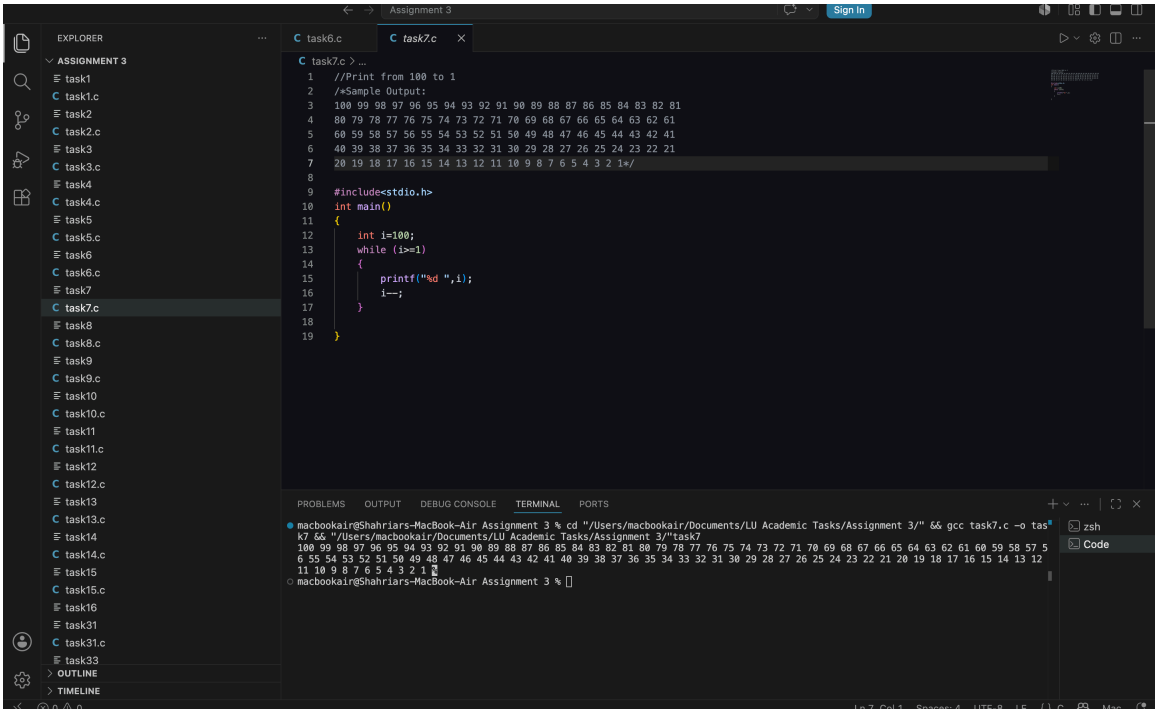


7. Print numbers from 100 to 1. Using while loop

Program:

```c
#include<stdio.h>
int main()
{
    int i=100;
    while (i>=1)
    {
        printf("%d ",i);
        i--;
    }
}
```

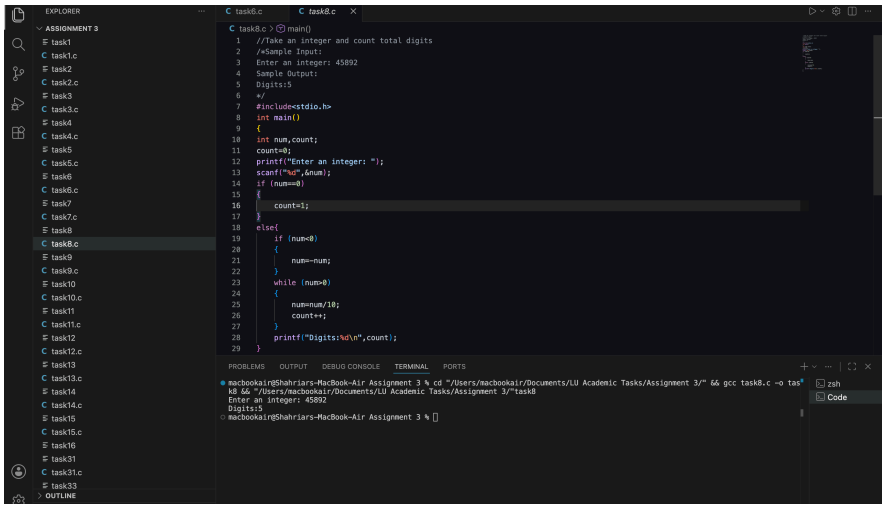
Output Screenshot:



8. Take an integer and count total digits.

Sample Input:
```
45892
```
Sample Output:
```
Total Digits = 5
```

Program:
```c
#include<stdio.h>
int main()
{
int num,count;
count=0;
printf("Enter an integer: ");
scanf("%d",&num);
if (num==0)
{
    count=1;
}
else{
    if (num<0)
    {
        num=-num;
    }
    while (num>0)
    {
        num=num/10;
        count++;
    }
    printf("Digits:%d\n",count);
}
}
```
Output Screenshot:

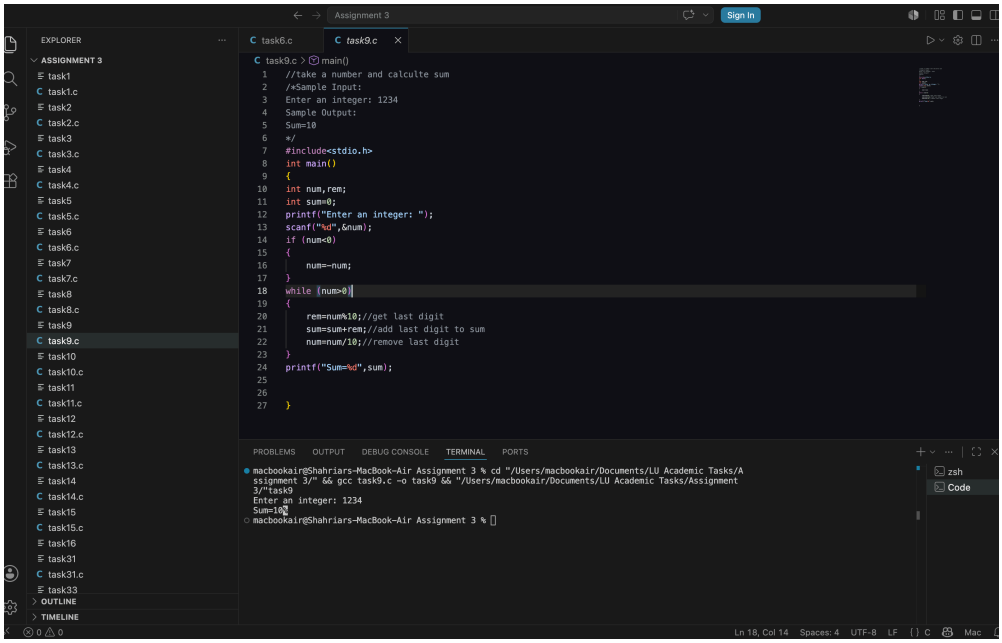


9. Take a number and calculate sum of all digits.

Sample Input:
```
1234
```
Sample Output:
```
Sum = 10
```

Program:
```
#include<stdio.h>
int main()
{
int num,rem;
int sum=0;
printf("Enter an integer: ");
scanf("%d",&num);
if (num<0)
{
    num=-num;
}
while (num>0)
{
    rem=num%10;//get last digit
    sum=sum+rem;//add last digit to sum
    num=num/10;//remove last digit
}
printf("Sum=%d",sum);
}
```

Output Screenshot:



10. Take an integer and reverse it using loop.

Sample Input:

```
12345
```

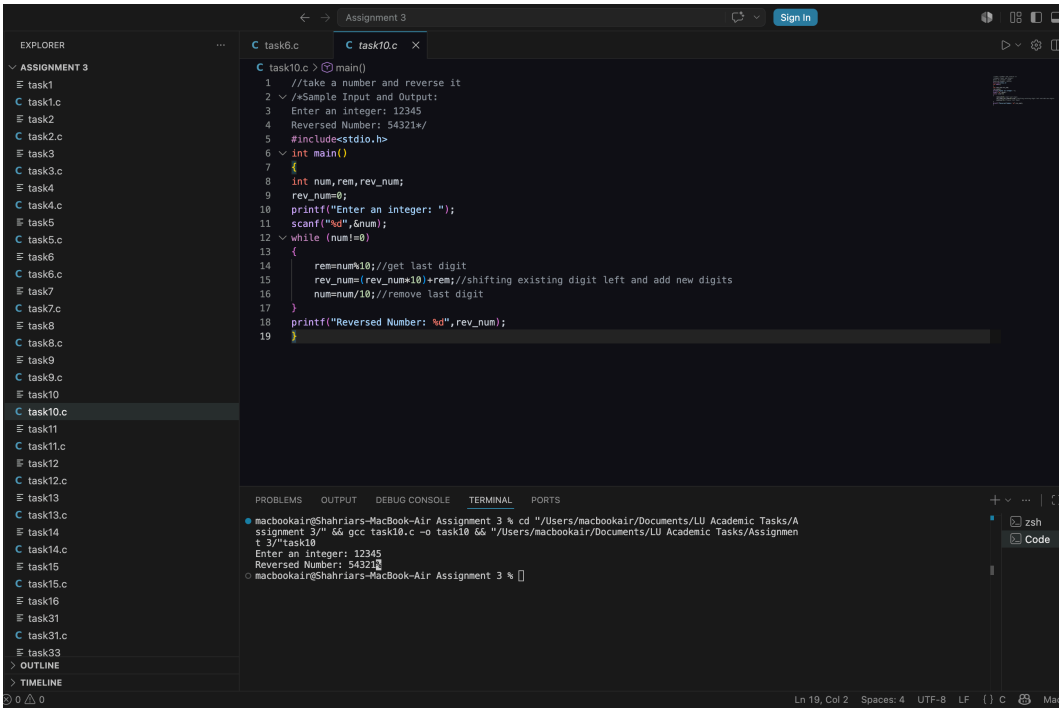
Sample Output:

```
54321
```

Program:

```
#include<stdio.h>
int main()
{
int num,rem,rev_num;
rev_num=0;
printf("Enter an integer: ");
scanf("%d",&num);
while (num!=0)
{
    rem=num%10;//get last digit
    rev_num=(rev_num*10)+rem;//shifting existing digit left and
add new digits
    num=num/10;//remove last digit
}
printf("Reversed Number: %d",rev_num);
}
```

Output Screenshot:



Section 3: Do-While Loop Problems

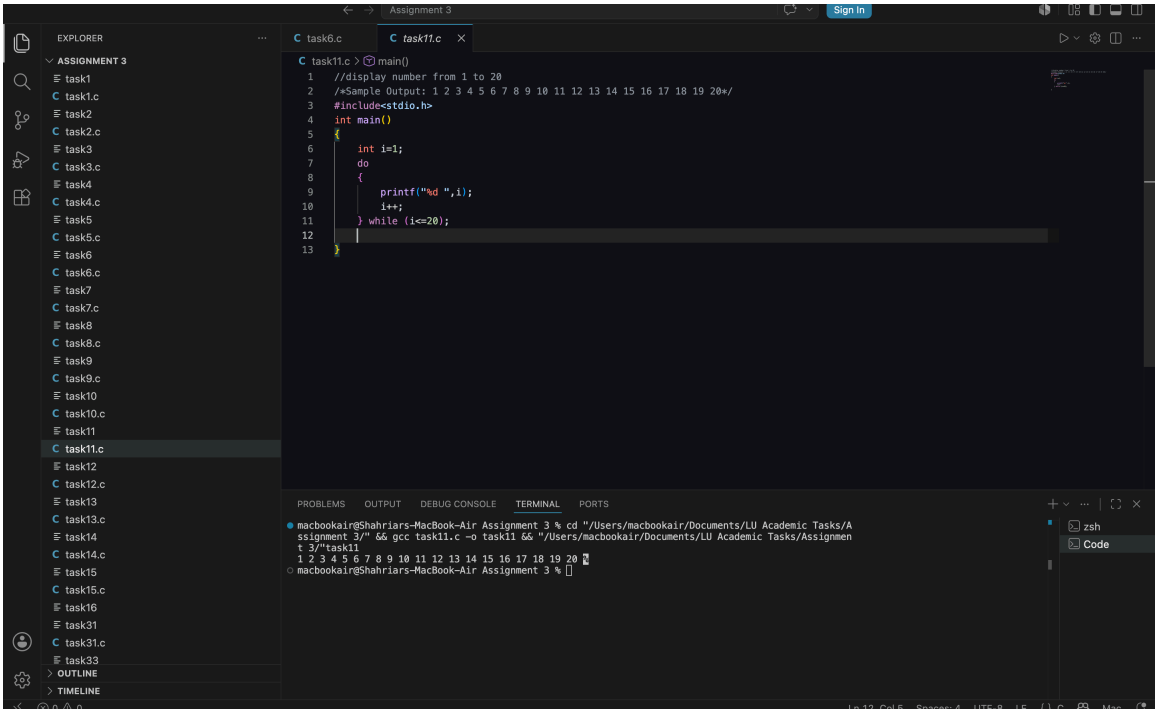
11. Display numbers from 1 to 20 using `do-while`.

Program:

```c
#include<stdio.h>
int main()
{
    int i=1;
    do
    {
        printf("%d ",i);
        i++;
    } while (i<=20);

}
```

Output Screenshot:



12. Create a menu driven calculator using `do-while`.

Operations:

```
1. Addition
2. Subtraction
3. Multiplication
4. Division
5. Exit
```

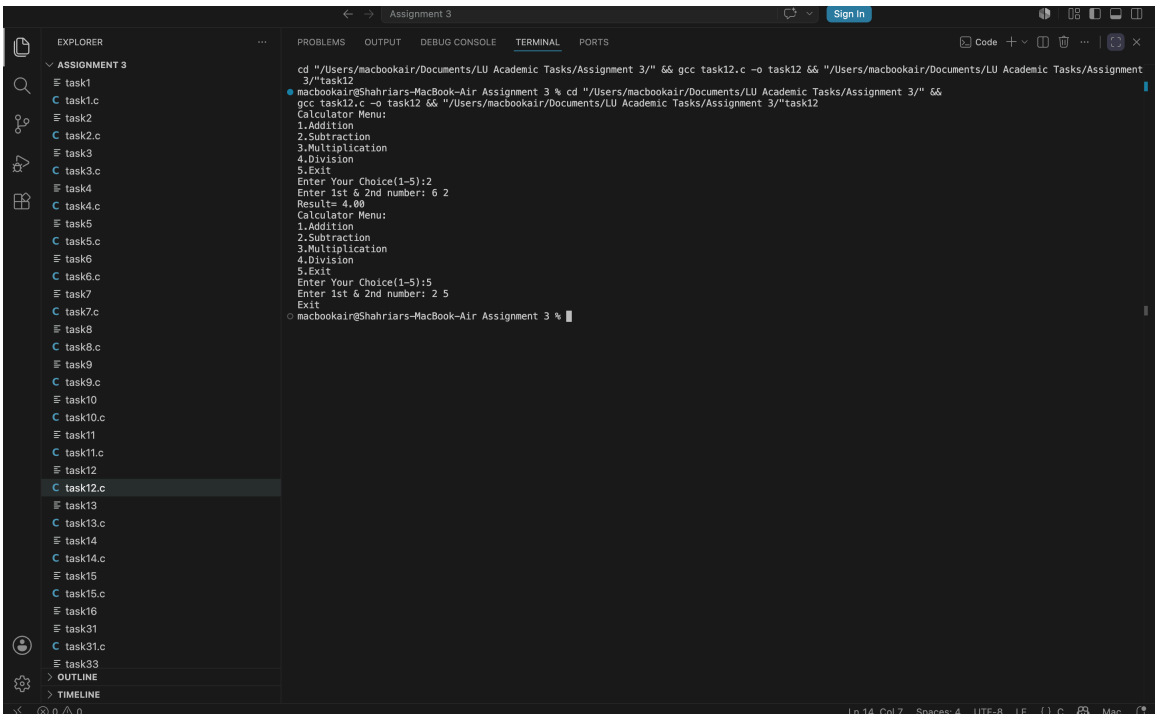
Program:

```c
#include<stdio.h>
int main()
{
    int choice;
    double num1,num2;
    do
    {
        printf("Calculator Menu:
\n1.Addition\n2.Subtraction\n3.Multiplication\n4.Division\n5.Exi
t\n");
        printf("Enter Your Choice(1-5):");
        scanf("%d",&choice);
        printf("Enter 1st & 2nd number: ");
        scanf("%lf %lf",&num1,&num2);

    switch(choice)
    {
    case 1:
        printf("Result= %.2lf\n",num1+num2);
        break;
    case 2:
        printf("Result= %.2lf\n",num1-num2);
        break;
    case 3:
        printf("Result= %.2lf\n",num1*num2);
        break;
    case 4:
        printf("Result= %.2lf\n",num1/num2);
        break;
    case 5:
        printf("Exit\n");
        break;
    default:
        printf("Invalid!\n");
        break;
    }
```

```
    }
    while (choice!=5);
}
```

Output Screenshot:



13. Keep taking password input until correct password is entered. If the password is wrong then
display 'Incorrect password! Please try again!'
Assume Correct Password is: `1234`

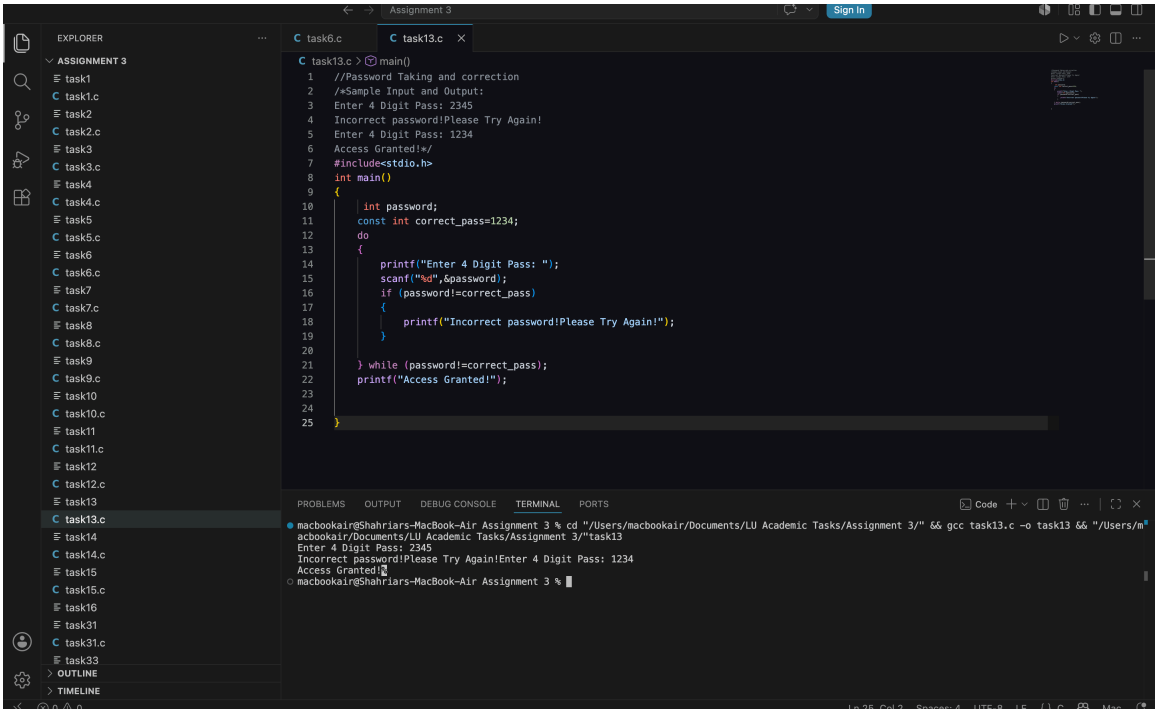
Program:

```
#include<stdio.h>
int main()
{
     int password;
    const int correct_pass=1234;
    do
    {
        printf("Enter 4 Digit Pass: ");
        scanf("%d",&password);
```

```
        if (password!=correct_pass)
        {
            printf("Incorrect password!Please Try Again!");
        }

    } while (password!=correct_pass);
    printf("Access Granted!");
}
```

Output Screenshot:



14. User keeps guessing until correct number is entered. Suppose the correct number is 7. If the user guess any incorrect number then display ‘wrong guess!’

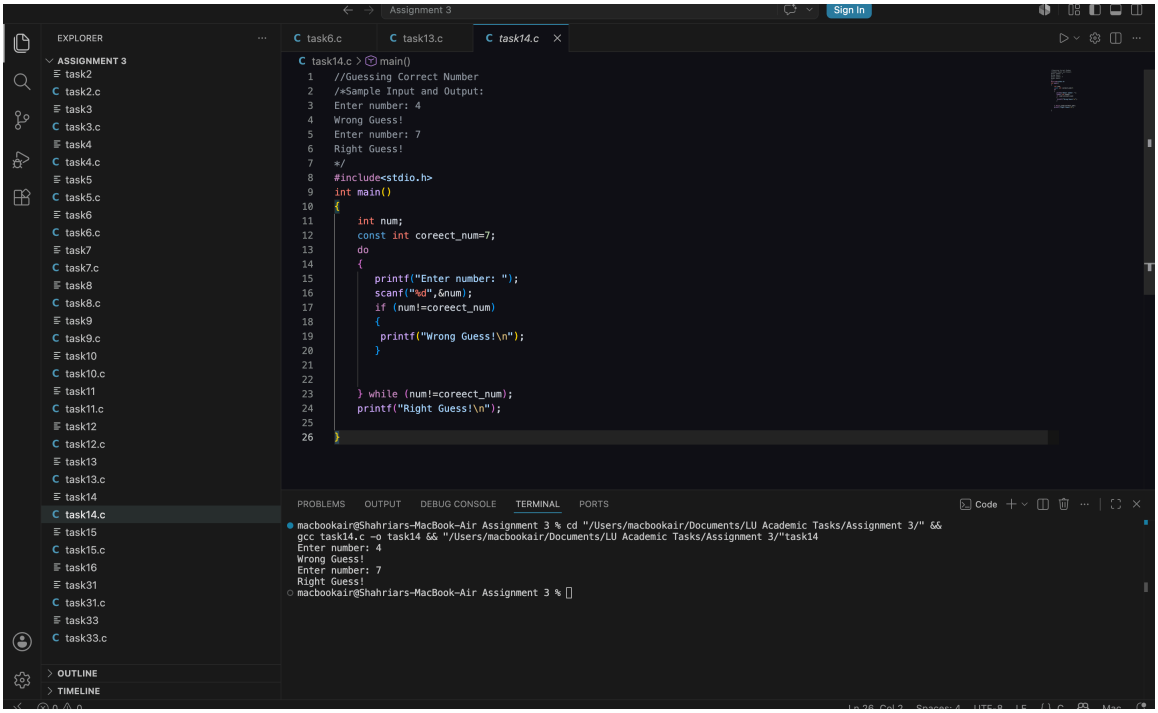
Program:

```
#include<stdio.h>
int main()
{
    int num;
    const int coreect_num=7;
    do
```

```
    {
        printf("Enter number: ");
        scanf("%d",&num);
        if (num!=coreect_num)
        {
         printf("Wrong Guess!\n");
        }
    } while (num!=coreect_num);
    printf("Right Guess!\n");

}
```

Output Screenshot:



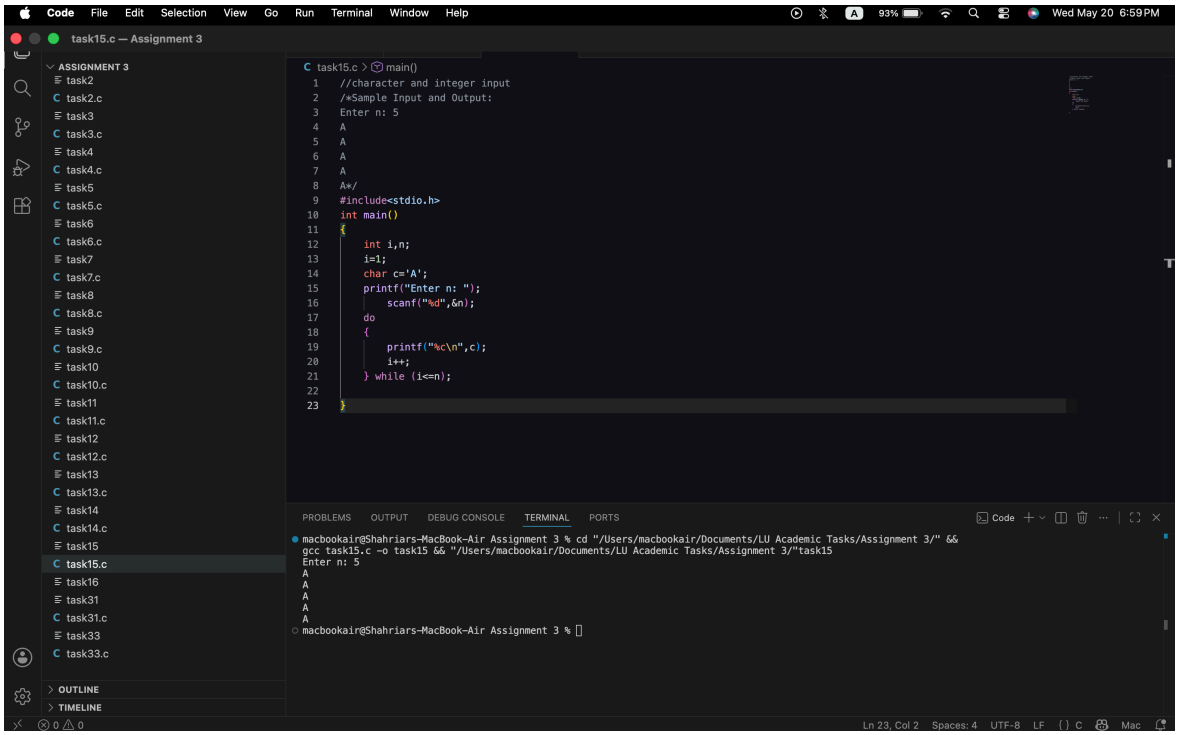
15. Write a program that takes a character and an integer N as input, then prints the given character N times using a do-while loop.

Program:

```
#include<stdio.h>
int main()
{
    int i,n;
```

```
    i=1;
    char c='A';
    printf("Enter n: ");
        scanf("%d",&n);
    do
    {
        printf("%c\n",c);
        i++;
    } while (i<=n);
}
```

Output Screenshot:



Section 4: Nested Loop Fundamentals

16. Square Star Pattern

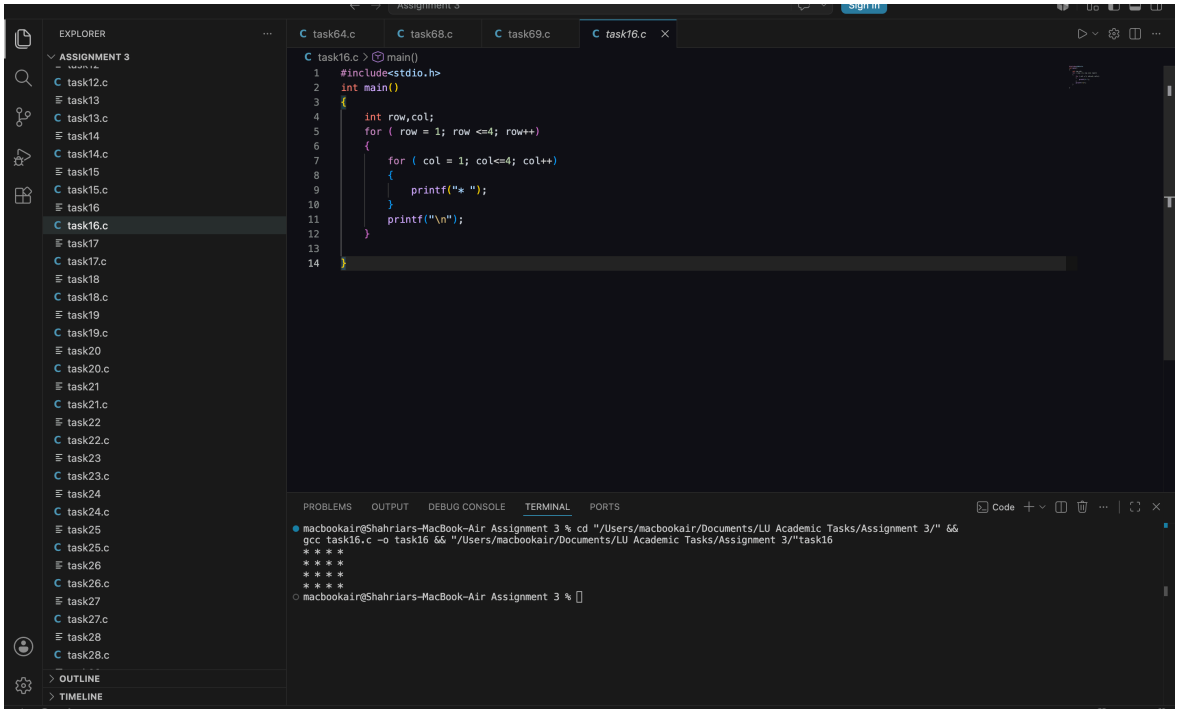
Print:

```
* * * *
* * * *
* * * *
* * * *
```

Program:

```
#include<stdio.h>
int main()
{
    int row,col;
    for ( row = 1; row <=4; row++)
    {
        for ( col = 1; col<=4; col++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output Screenshot:



17. Number Grid Pattern

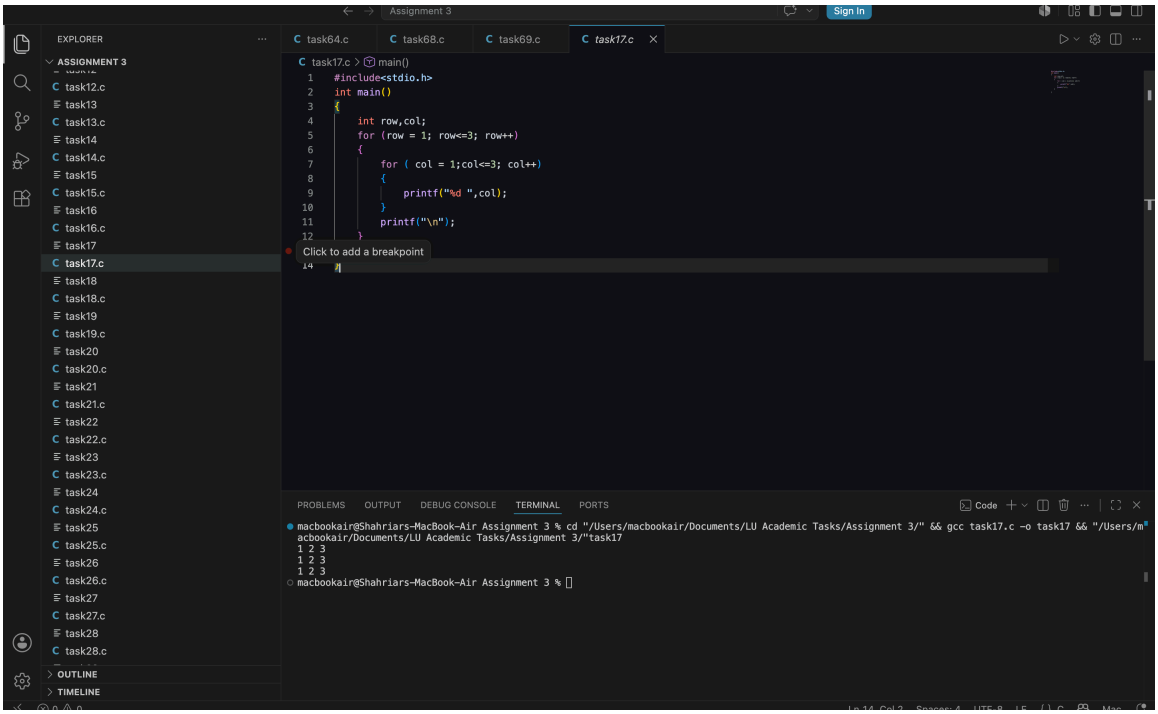
Print:

```
1 2 3
1 2 3
1 2 3
```

Program:

```c
#include<stdio.h>
int main()
{
    int row,col;
    for (row = 1; row<=3; row++)
    {
        for ( col = 1;col<=3; col++)
        {
            printf("%d ",col);
        }
        printf("\n");
    }
}
```

Output Screenshot:



18. Increasing Number Pattern

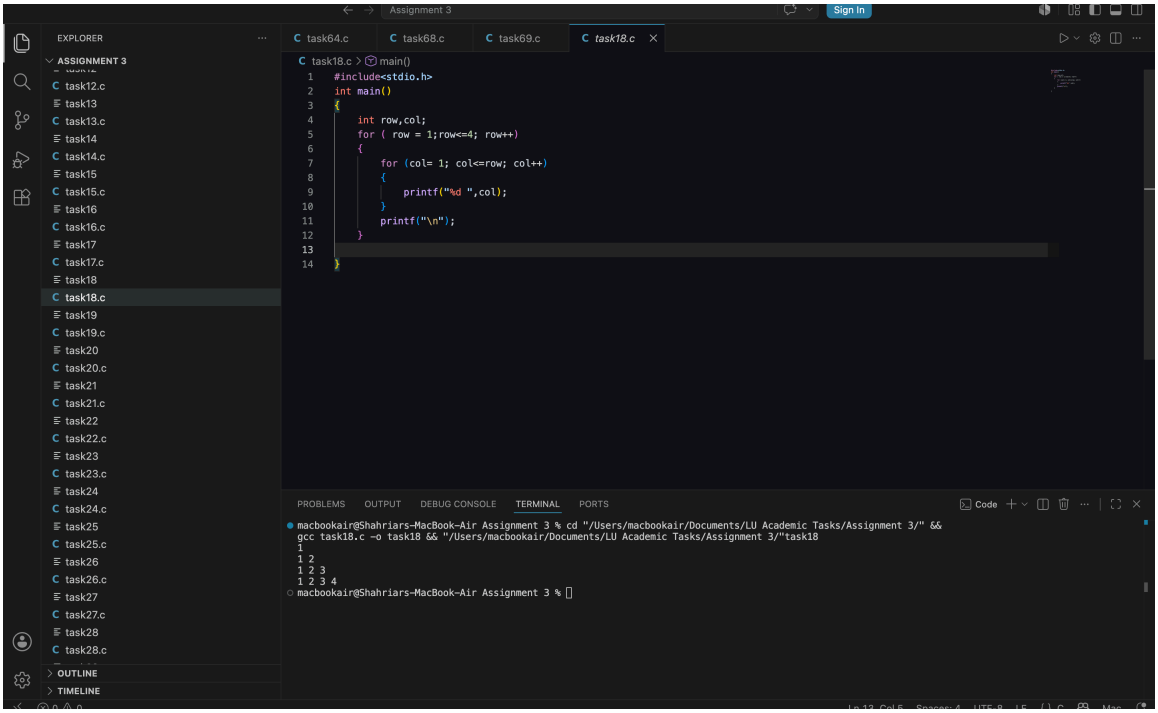
Print:

```
1
1 2
1 2 3
1 2 3 4
```

Program:

```c
#include<stdio.h>
int main()
{
    int row,col;
    for ( row = 1;row<=4; row++)
    {
        for (col= 1; col<=row; col++)
        {
            printf("%d ",col);
        }
        printf("\n");
    }
}
```

Output Screenshot:



19. Triangle Star Pattern

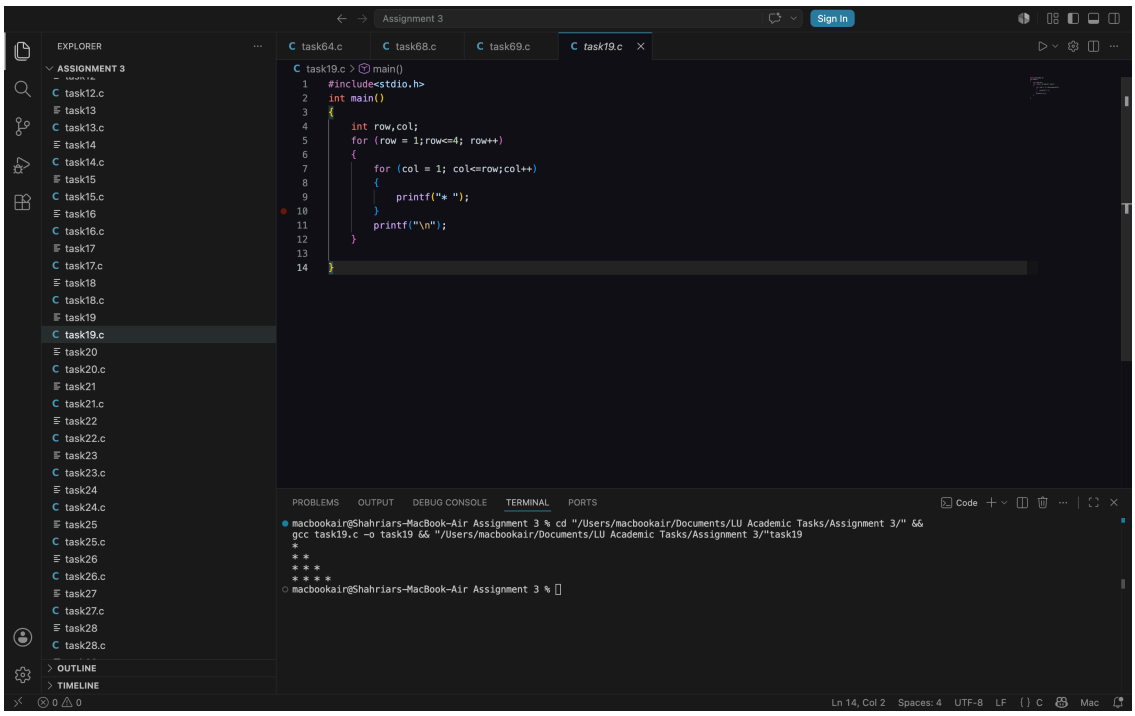
Print:

```
*
* *
* * *
* * * *
```

Program:

```
#include<stdio.h>
int main()
{
    int row,col;
    for (row = 1;row<=4; row++)
    {
        for (col = 1; col<=row;col++)
        {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output Screenshot:



20. Reverse Triangle Pattern

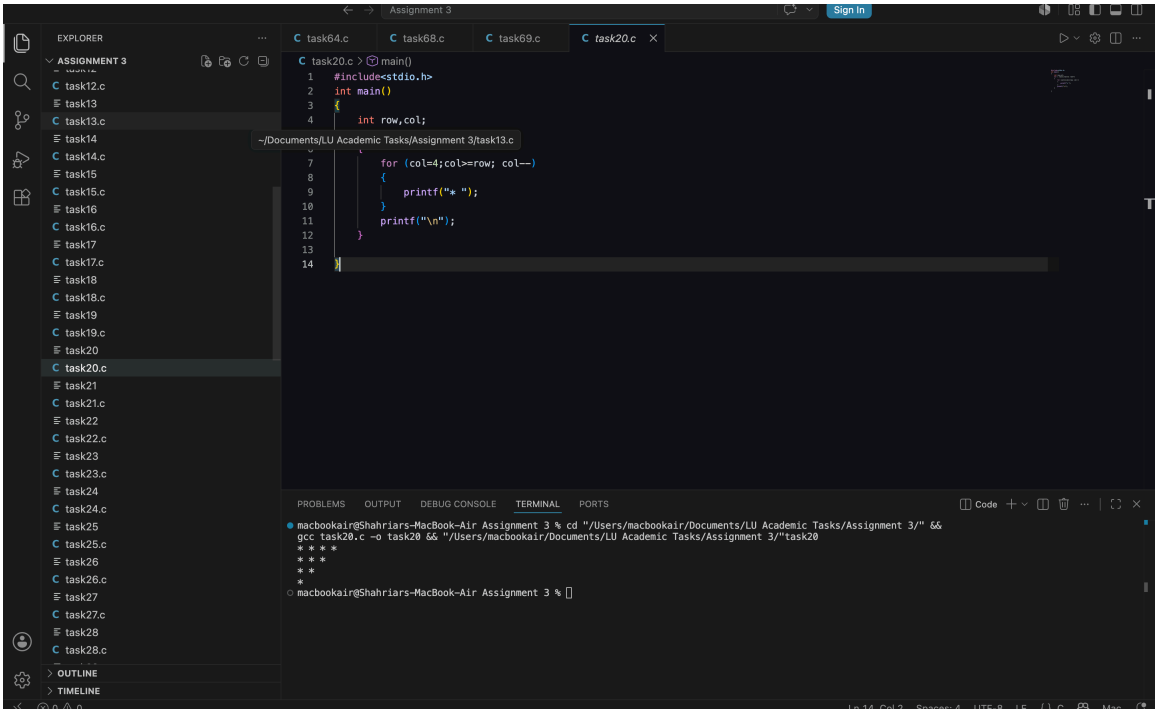
Print:

```
* * * *
* * *
* *
*
```

Program:

```
#include<stdio.h>
int main()
{
    int row,col;
    for ( row=1;row<=4; row++)
    {
        for (col=4;col>=row; col--)
        {
            printf("* ");
        }
        printf("\n");
    }
}
```

Output Screenshot:



Section 5: Advanced Pattern Printing

21. Right Aligned Star Triangle

Print:

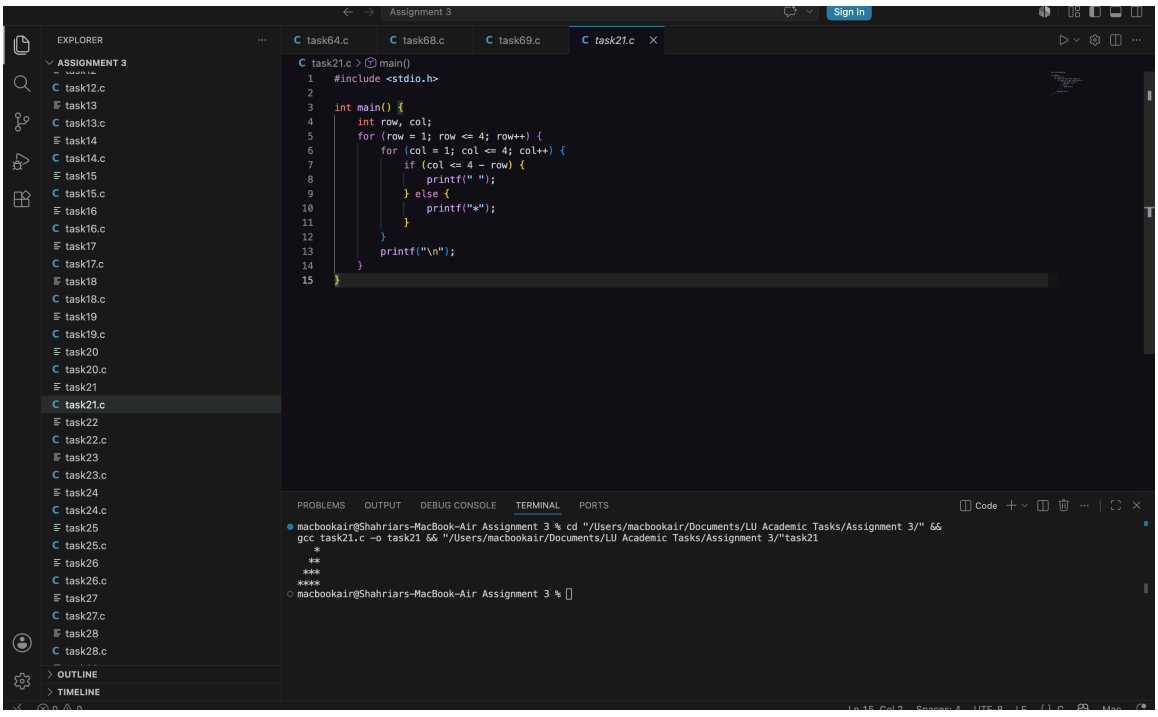
```
      *
    * *
  * * *
* * * *
```

Program:

```c
#include <stdio.h>

int main() {
    int row, col;
    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= 4; col++) {
            if (col <= 4 - row) {
                printf(" ");
            } else {
                printf("*");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



22. Pyramid Pattern

Print:

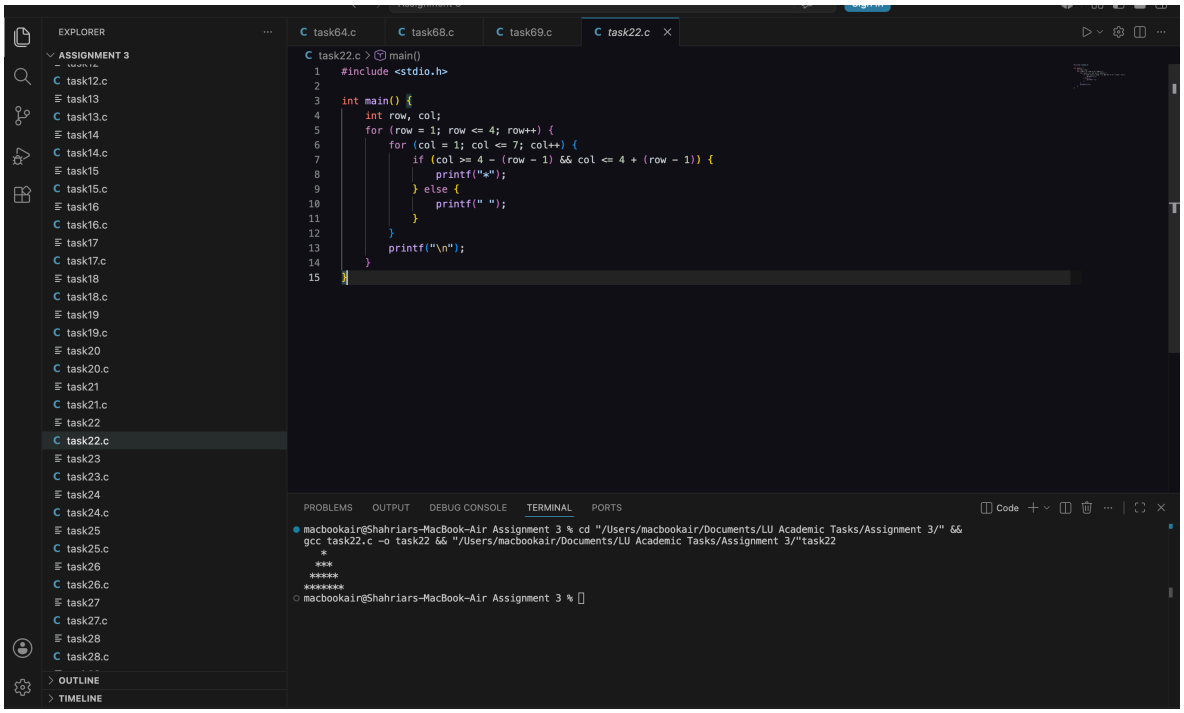
```
      *
    * * *
  * * * * *
* * * * * * *
```

Program:

```c
#include <stdio.h>

int main() {
    int row, col;
    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= 7; col++) {
            if (col >= 4 - (row - 1) && col <= 4 + (row - 1)) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



23. Inverted Pyramid

Print:

```
* * * * * * *
 * * * * * *
   * * * *
     *
```

Program:

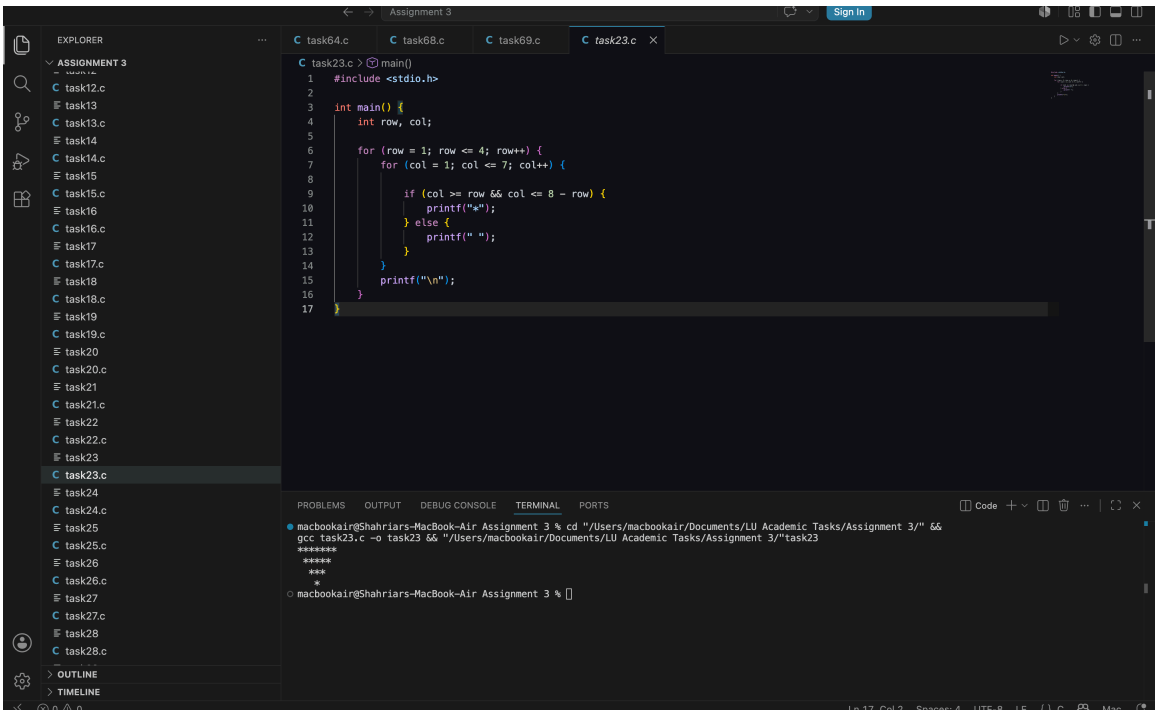
```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= 7; col++) {

            if (col >= row && col <= 8 - row) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



24. Floyd's Triangle

Print:

```
1
2 3
4 5 6
7 8 9 10
```

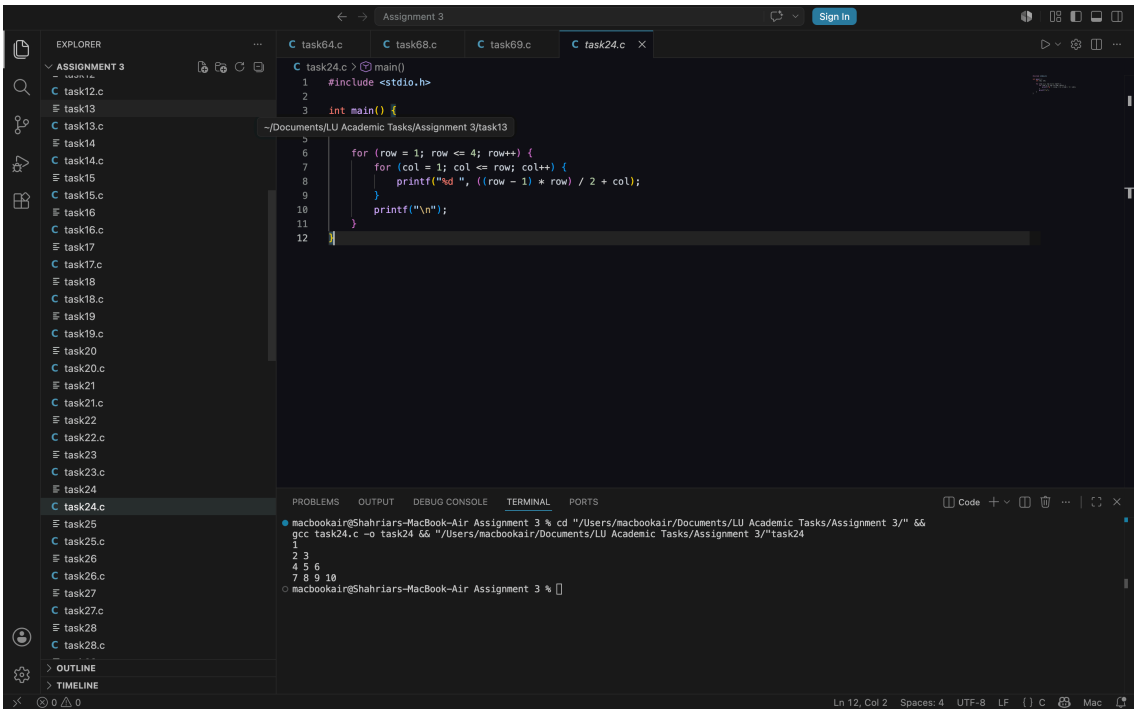
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= row; col++) {
            printf("%d ", ((row - 1) * row) / 2 + col);
        }
        printf("\n");
    }
}
```

Output Screenshot:



25. Alphabet Triangle

Print:

```
A
A B
A B C
A B C D
```

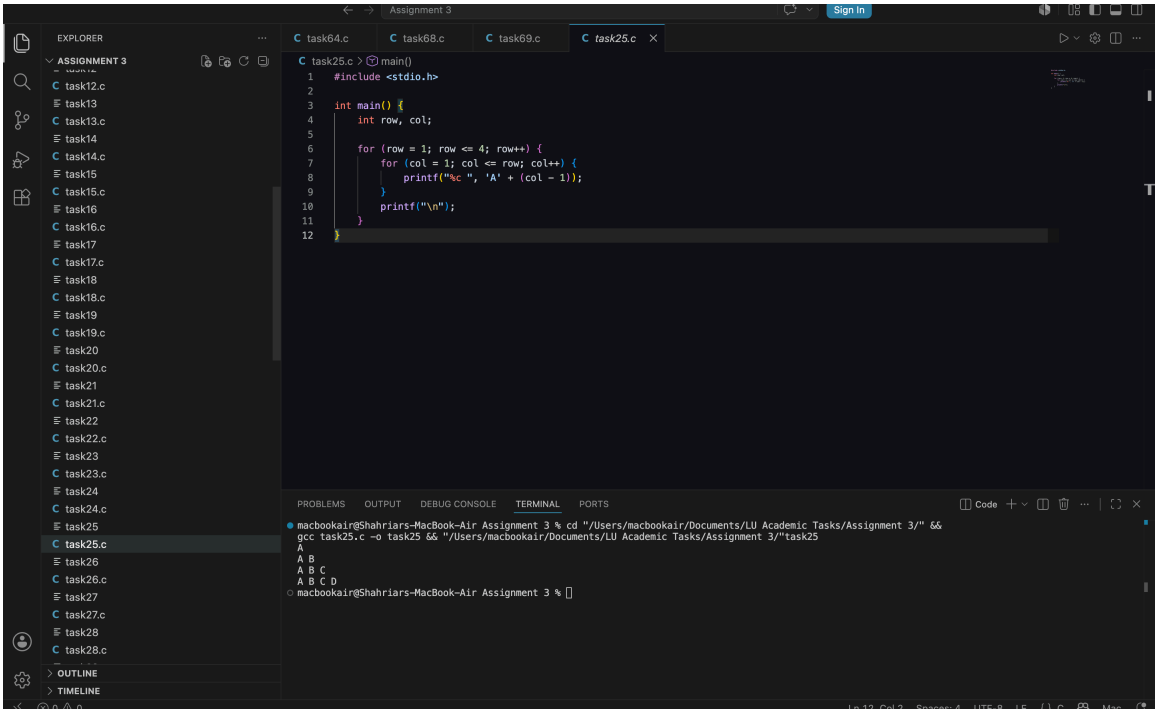
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= row; col++) {
            printf("%c ", 'A' + (col - 1));
        }
        printf("\n");
    }
}
```

Output Screenshot:



26. Binary Pattern

Print:

```
1
0 1
0 1 0
1 0 1 0
```

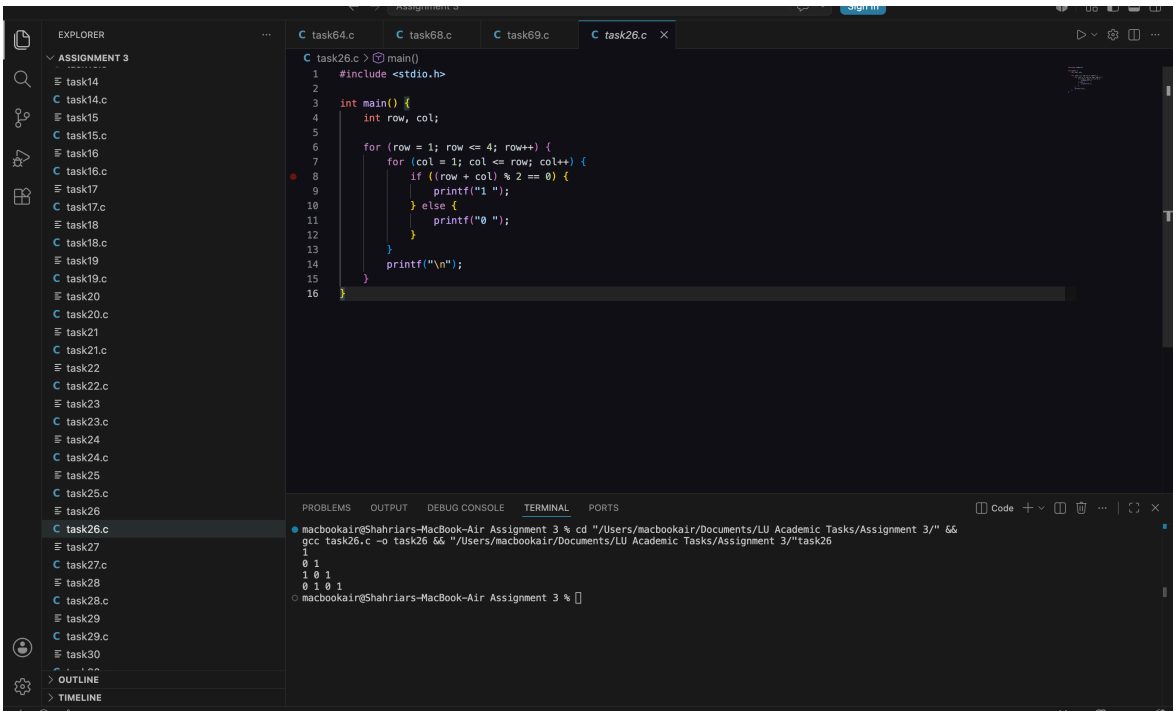
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 4; row++) {
        for (col = 1; col <= row; col++) {
            if ((row + col) % 2 == 0) {
                printf("1 ");
            } else {
                printf("0 ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



27. Hollow Square Pattern

Print:

```
* * * * *
*       *
*       *
*       *
* * * * *
```

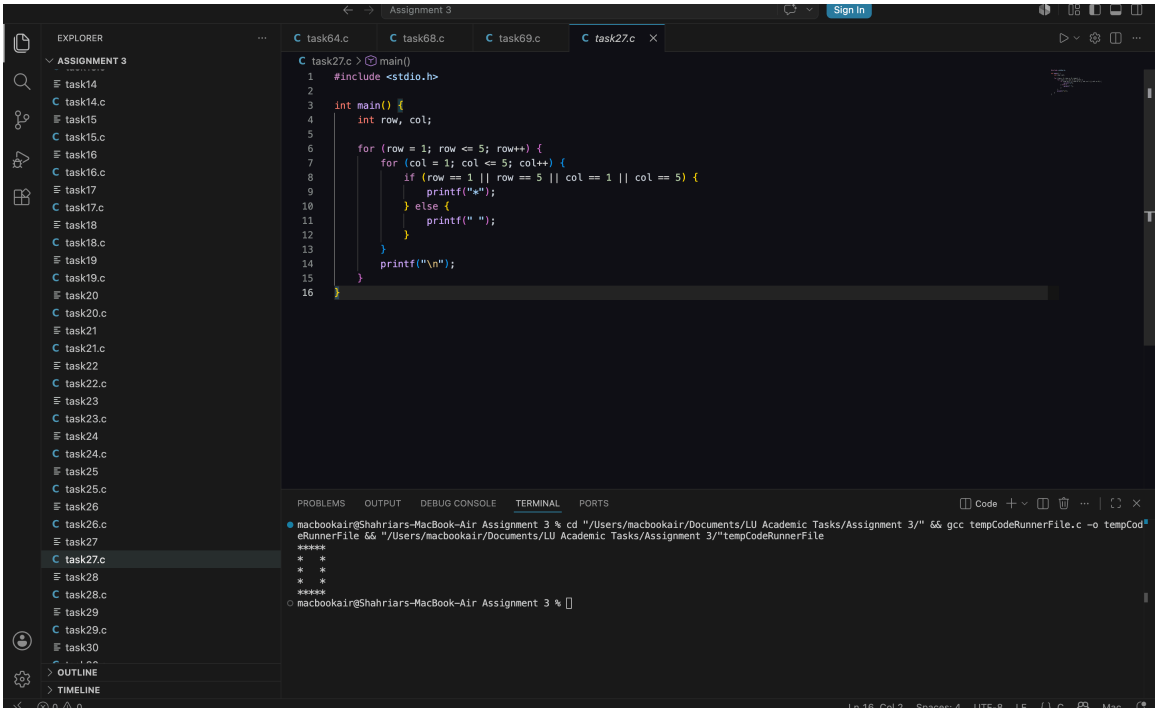
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 5; row++) {
        for (col = 1; col <= 5; col++) {
            if (row == 1 || row == 5 || col == 1 || col == 5) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



28. Create a hollow pyramid using nested loops.(Try)
Print:

```
    *
   * *
  *   *
 *     *
*********
```

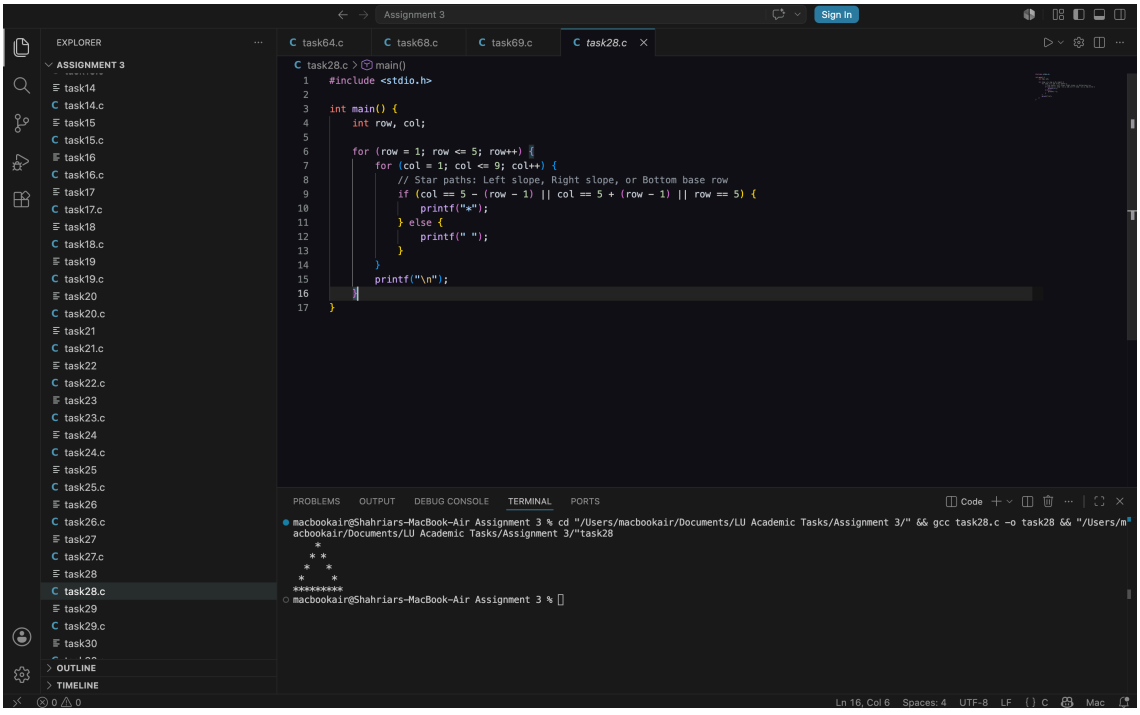
Program:

```
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 5; row++) {
        for (col = 1; col <= 9; col++) {
            // Star paths: Left slope, Right slope, or Bottom base row
            if (col == 5 - (row - 1) || col == 5 + (row - 1) || row == 5) {
                printf("*");
            } else {
                printf(" ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



29. Print diamond shape using stars.(Try)

Print:

```
    *
   ***
  *****
 *******
*********
 *******
  *****
   ***
    *
```

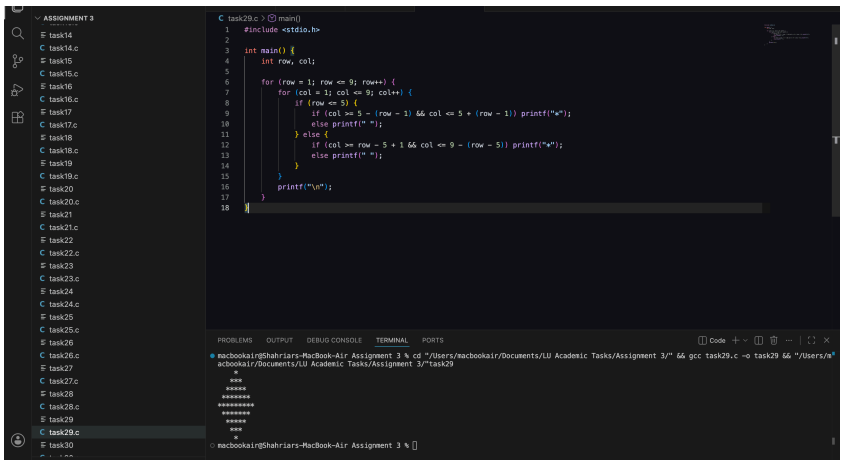
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 9; row++) {
        for (col = 1; col <= 9; col++) {
            if (row <= 5) {
                if (col >= 5 - (row - 1) && col <= 5 + (row - 1)) printf("*");
                else printf(" ");
            } else {
                if (col >= row - 5 + 1 && col <= 9 - (row - 5)) printf("*");
                else printf(" ");
            }
        }
        printf("\n");
    }
}
```

Output Screenshot:



30. Create butterfly star pattern.(Try)

Print:

```
* *
** **
*** ***
********
********
*** ***
** **
* *
```

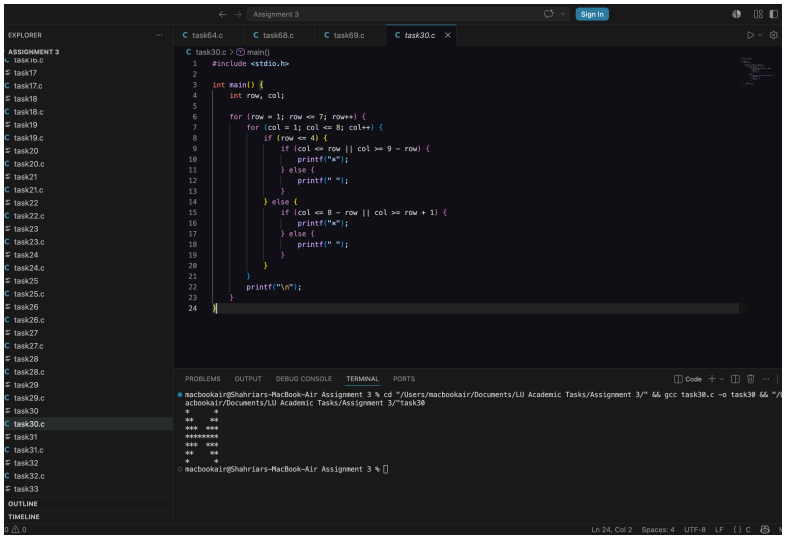
Program:

```c
#include <stdio.h>

int main() {
    int row, col;

    for (row = 1; row <= 7; row++) {
        for (col = 1; col <= 8; col++) {
            if (row <= 4) {
                if (col <= row || col >= 9 - row) {
                    printf("*");
                } else {
                    printf(" ");
                }
            } else {
                if (col <= 8 - row || col >= row + 1) {
                    printf("*");
                } else {
                    printf(" ");}}}
        printf("\n");}
}
```

Output Screenshot:



Section 6: Loop with Mathematical Applications

31. Calculate factorial using loop (n!).

Sample Input:

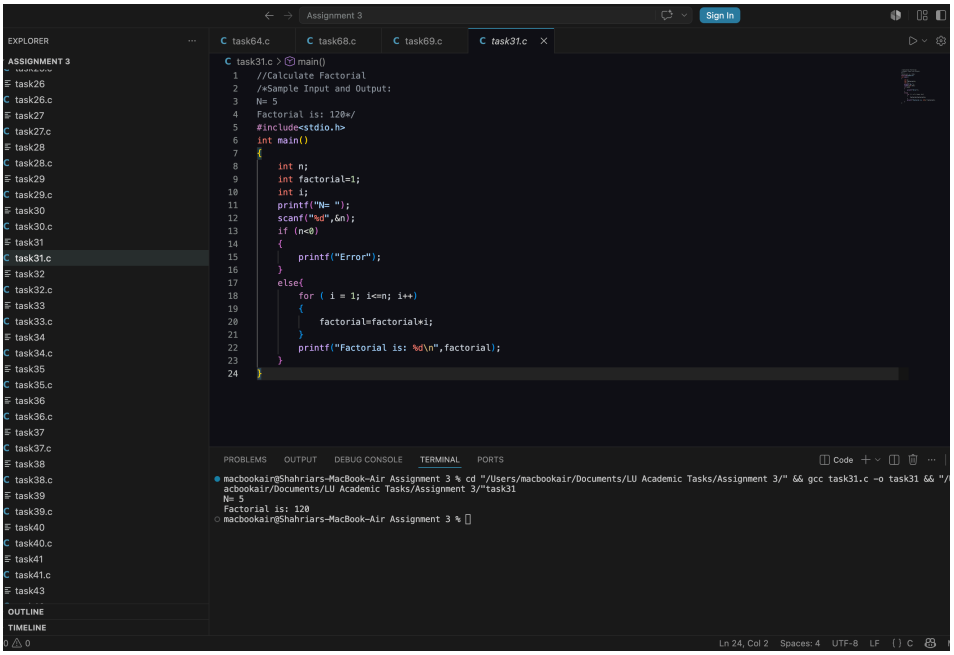
Sample Output:

```
5
120
```

Program:

```c
#include<stdio.h>
int main()
{
    int n;
    int factorial=1;
    int i;
    printf("N= ");
    scanf("%d",&n);
    if (n<0)
    {
        printf("Error");
    }
    else{
        for ( i = 1; i<=n; i++)
        {
            factorial=factorial*i;
        }
        printf("Factorial is: %d\n",factorial);
    }
}
```

Output Screenshot:



32. Calculate: `base^power` using loop.

Sample Input:

Sample Output:

```
Base = 2
Power = 4
16
```

Program:

```c
#include <stdio.h>

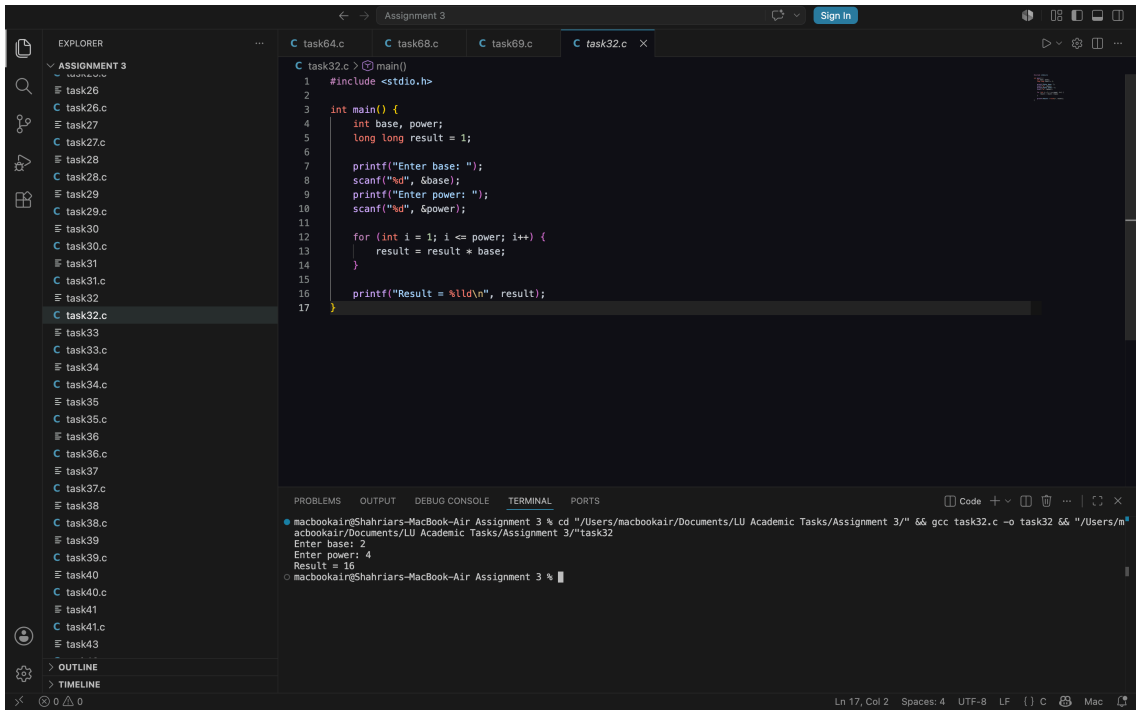
int main() {
    int base, power;
    long long result = 1;

    printf("Enter base: ");
    scanf("%d", &base);
    printf("Enter power: ");
    scanf("%d", &power);

    for (int i = 1; i <= power; i++) {
        result = result * base;
    }

    printf("Result = %lld\n", result);
}
```

Output Screenshot:



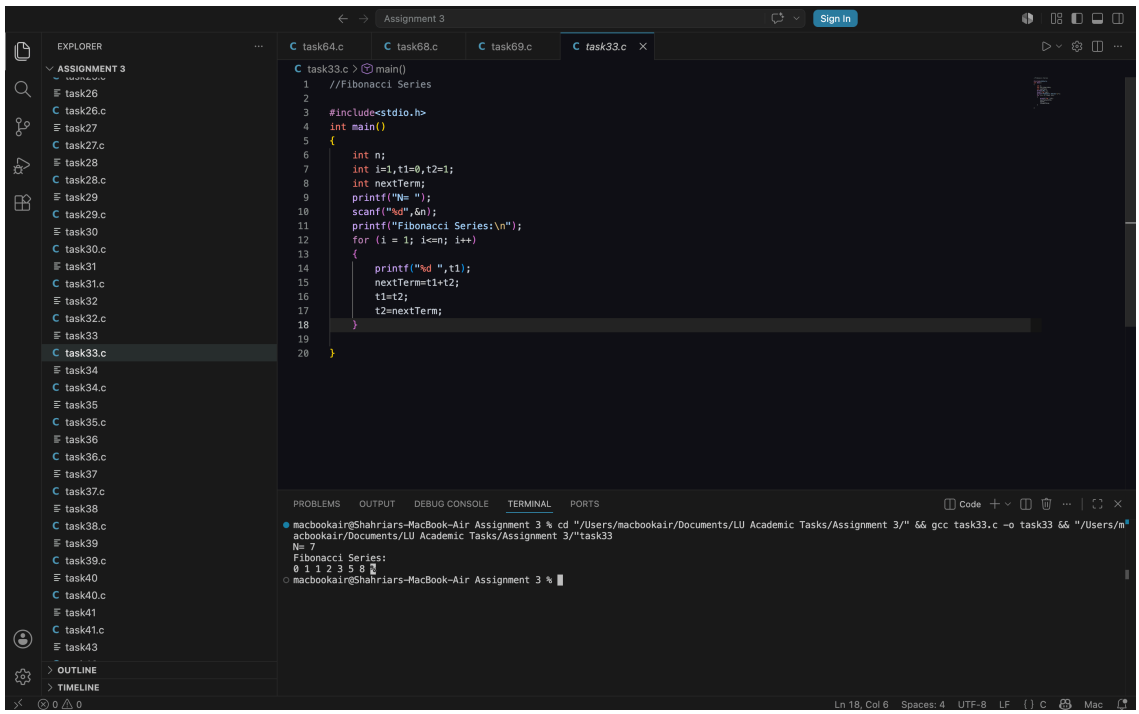
33. Print Fibonacci series up to N terms.
Sample Input: 7
Sample Output:

```
0 1 1 2 3 5 8
```

Program:

```c
#include<stdio.h>
int main()
{
    int n;
    int i=1,t1=0,t2=1;
    int nextTerm;
    printf("N= ");
    scanf("%d",&n);
    printf("Fibonacci Series:\n");
    for (i = 1; i<=n; i++)
    {
        printf("%d ",t1);
        nextTerm=t1+t2;
        t1=t2;
        t2=nextTerm;
    }   }
```

Output Screenshot:



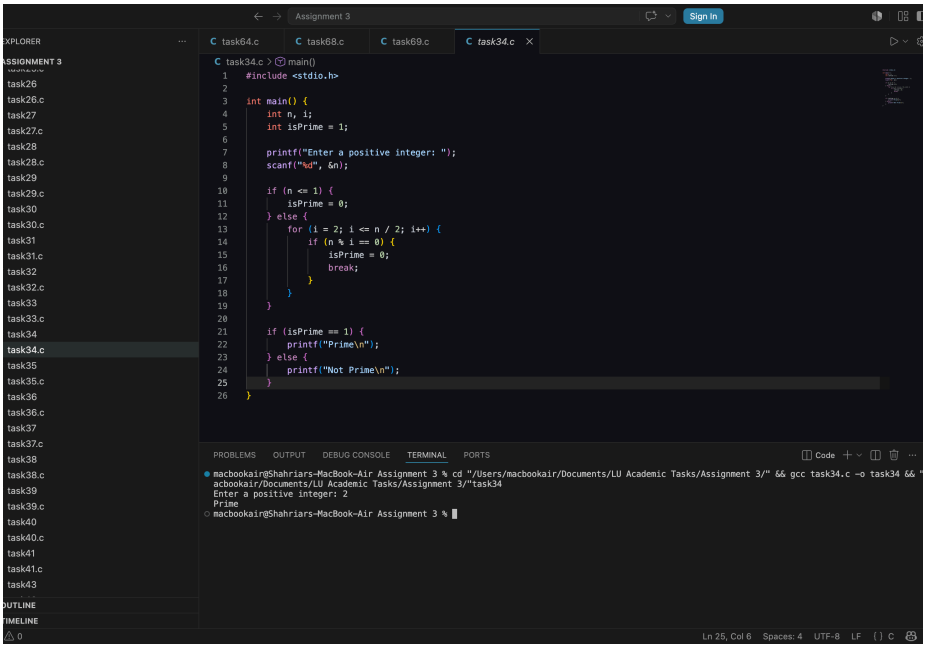
34. Check whether a number is prime or not using loop.
Sample Input: 2 4
Sample Output:

```
Prime
Not Prime
```

Program:

```c
#include <stdio.h>
int main() {
    int n, i;
    int isPrime = 1;
    printf("Enter a positive integer: ");
    scanf("%d", &n);
    if (n <= 1) {
        isPrime = 0;
    } else {
        for (i = 2; i <= n / 2; i++) {
            if (n % i == 0) {
                isPrime = 0;
                break;
            }
        }
    }
    if (isPrime == 1) {
        printf("Prime\n");
    } else {
        printf("Not Prime\n");}}
```

Output Screenshot:



35. Display all prime numbers from 1 to 100.

Program:

```c
#include <stdio.h>

int main() {
    int i, j, isPrime;

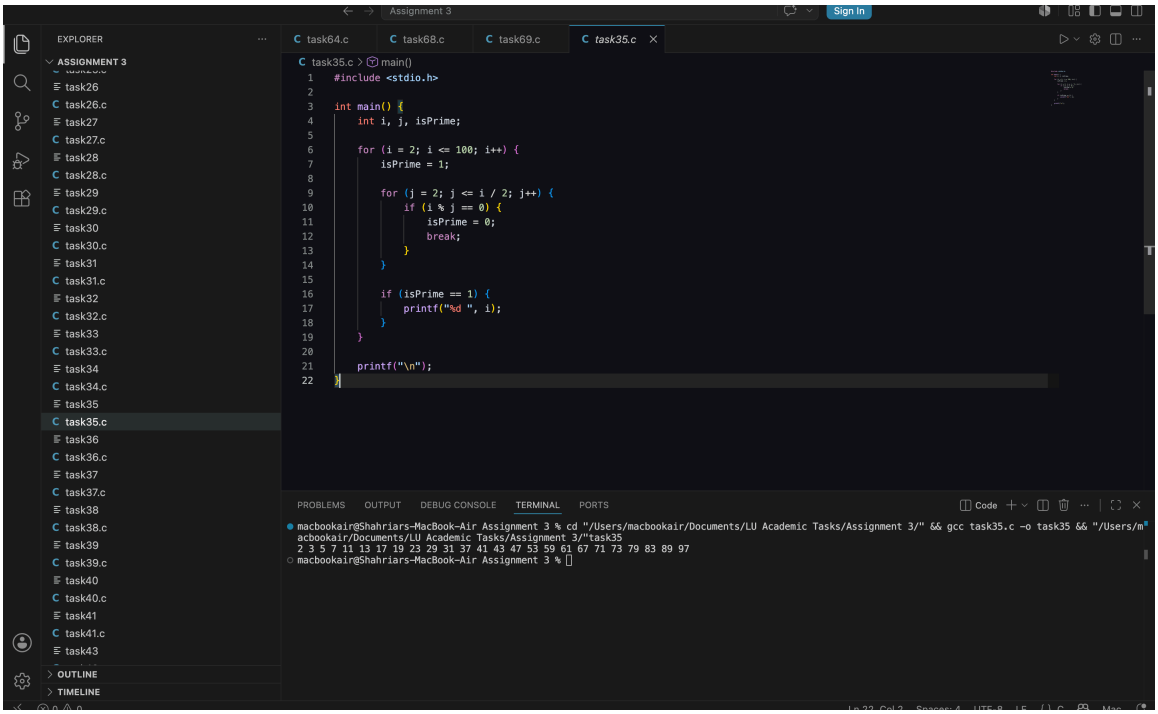
    for (i = 2; i <= 100; i++) {
        isPrime = 1;

        for (j = 2; j <= i / 2; j++) {
            if (i % j == 0) {
                isPrime = 0;
                break;
            }
        }

        if (isPrime == 1) {
            printf("%d ", i);
        }
    }

    printf("\n");
}
```

Output Screenshot:



36. Check whether a number is Armstrong or not. 153 is an Armstrong number.
Example:
$153 = 1^3 + 5^3 + 3^3$

Program:

```c
#include <stdio.h>

int main() {
    int n, originalNum, remainder, result = 0;

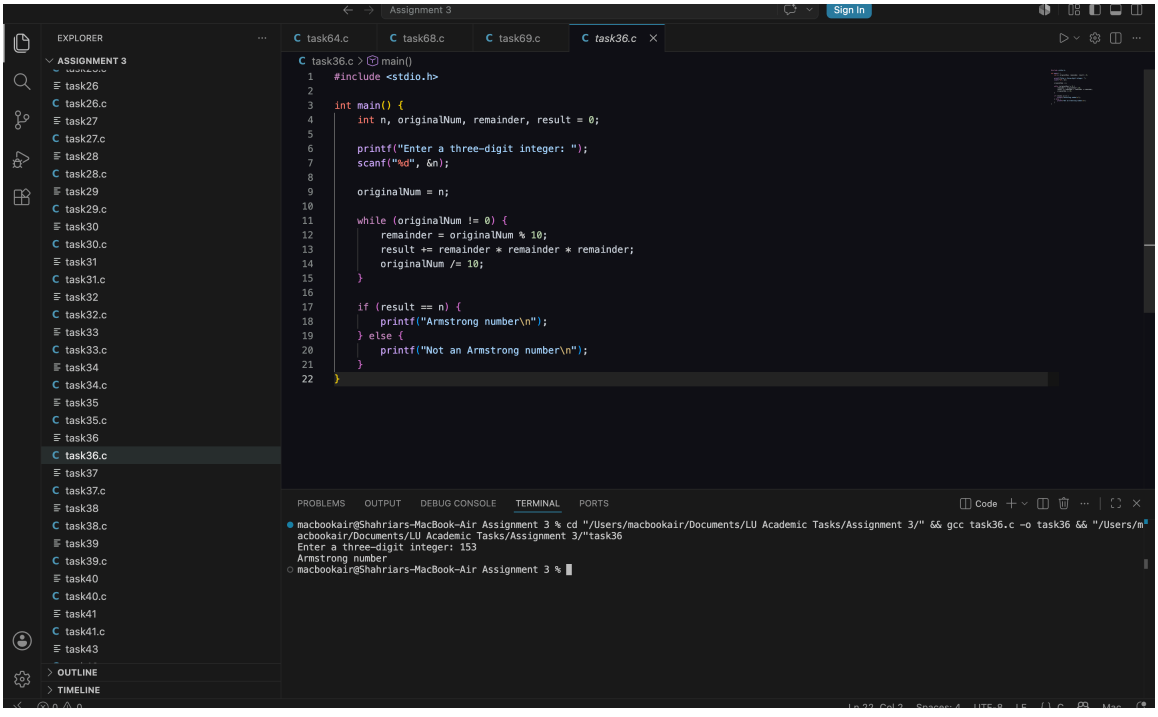
    printf("Enter a three-digit integer: ");
    scanf("%d", &n);

    originalNum = n;

    while (originalNum != 0) {
        remainder = originalNum % 10;
        result += remainder * remainder * remainder;
        originalNum /= 10;
    }

    if (result == n) {
        printf("Armstrong number\n");
    } else {
        printf("Not an Armstrong number\n");
    }
}
```

Output Screenshot:



37. Check whether a number is palindrome or Not.

Sample Input:

```
121
```

Sample Output:

```
Palindrome Number
```

Program:

```c
#include <stdio.h>

int main() {
    int n, originalNum, reversedNum = 0, remainder;

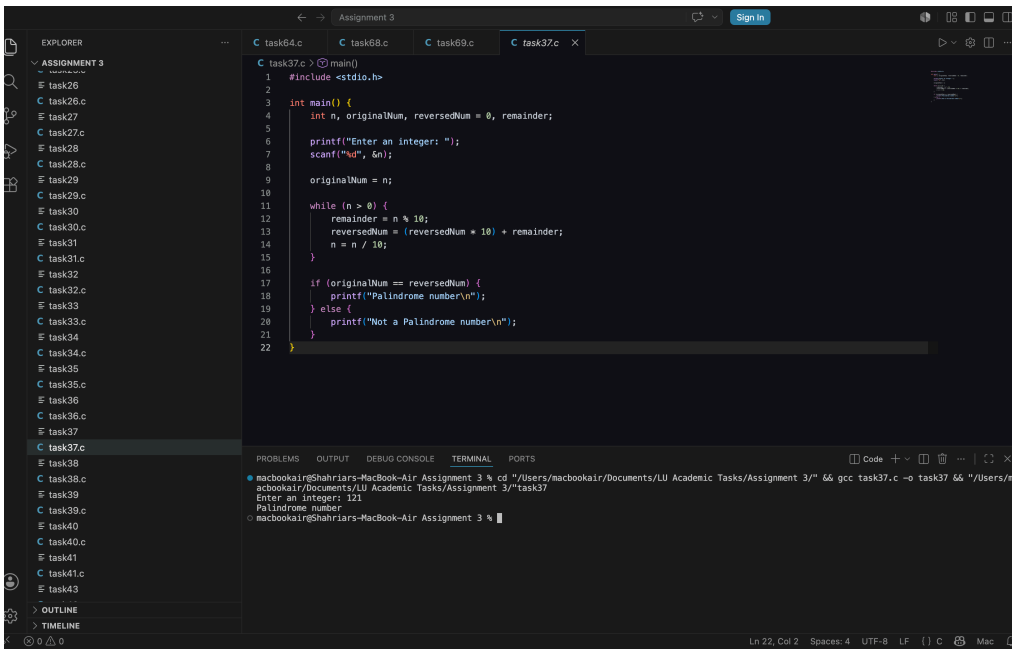
    printf("Enter an integer: ");
    scanf("%d", &n);

    originalNum = n;

    while (n > 0) {
        remainder = n % 10;
        reversedNum = (reversedNum * 10) + remainder;
        n = n / 10;
    }

    if (originalNum == reversedNum) {
        printf("Palindrome number\n");
    } else {
        printf("Not a Palindrome number\n");
    }
}
```

Output Screenshot:



38. Check whether a number is perfect or not. 6 is a perfect number.
Example:
```
6 = 1 + 2 + 3
```

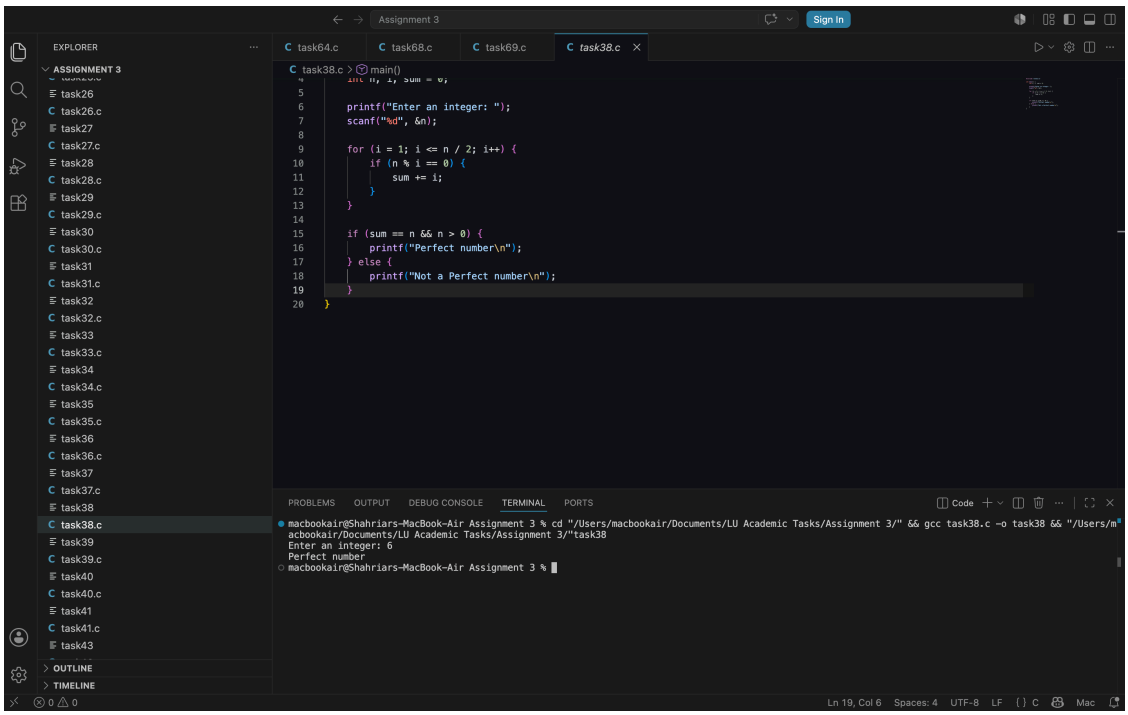
Program:
```c
#include <stdio.h>

int main() {
    int n, i, sum = 0;

    printf("Enter an integer: ");
    scanf("%d", &n);

    for (i = 1; i <= n / 2; i++) {
        if (n % i == 0) {
            sum += i;
        }
    }

    if (sum == n && n > 0) {
        printf("Perfect number\n");
    } else {
        printf("Not a Perfect number\n");
    }
}
```
Output Screenshot:



39. Take two number as input and Calculate their GCD(Greatest common divisor) and LCM(Least common multiple) using loops. (Try)

Program:

```c
#include <stdio.h>

int main() {
    int n1, n2, x, y, remainder, gcd, lcm;

    printf("Enter two integers: ");
    scanf("%d %d", &n1, &n2);

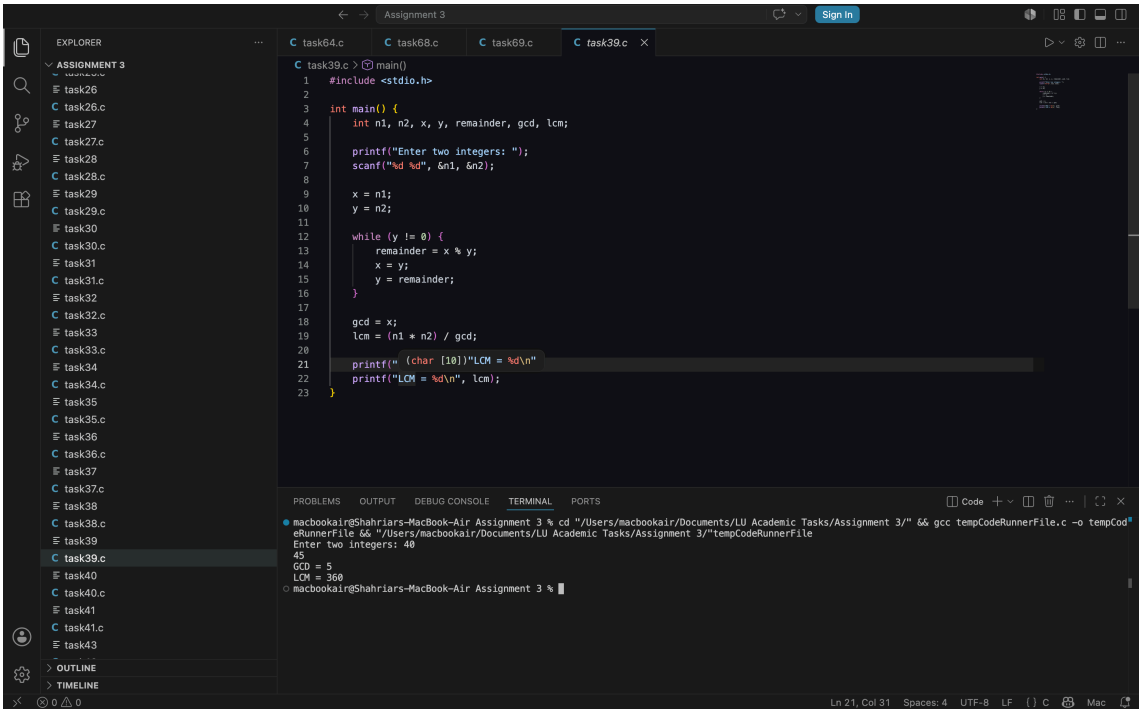
    x = n1;
    y = n2;

    while (y != 0) {
        remainder = x % y;
        x = y;
        y = remainder;
    }

    gcd = x;
    lcm = (n1 * n2) / gcd;

    printf("GCD = %d\n", gcd);
    printf("LCM = %d\n", lcm);
}
```

Output Screenshot:



40. Check whether a number is strong number.

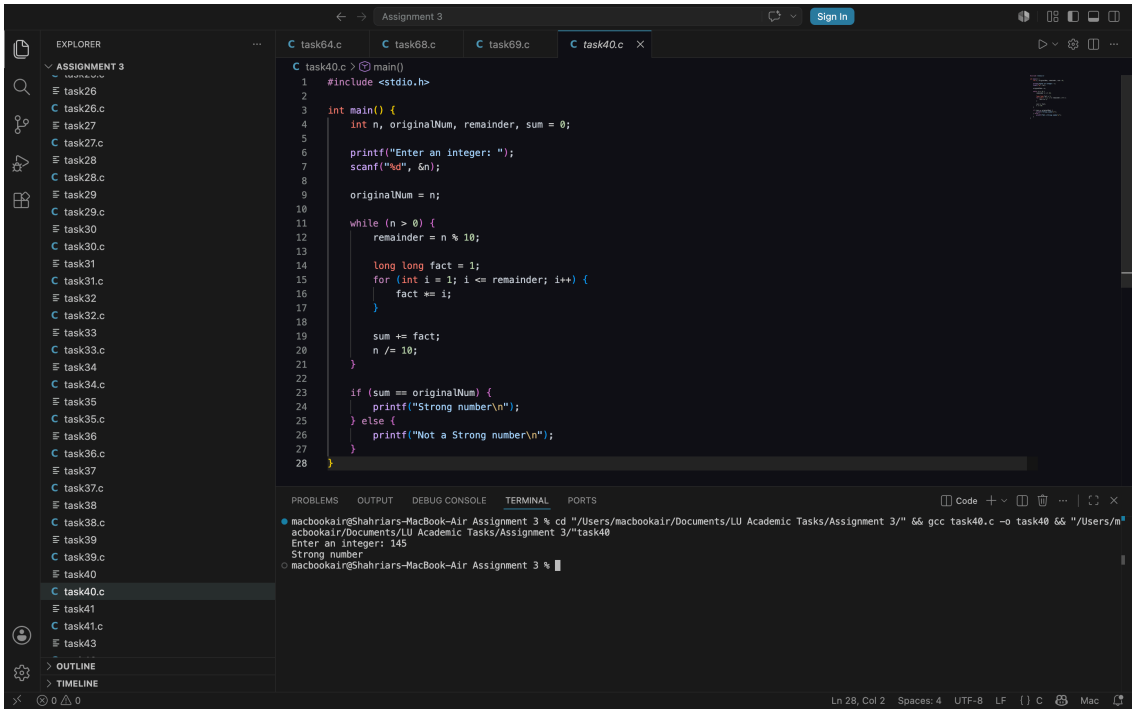
Example:

```
145 = 1! + 4! + 5!
```

Program:

```c
#include <stdio.h>
int main() {
    int n, originalNum, remainder, sum = 0;
    printf("Enter an integer: ");
    scanf("%d", &n);
    originalNum = n;
    while (n > 0) {
        remainder = n % 10;
        long long fact = 1;
        for (int i = 1; i <= remainder; i++) {
            fact *= i;
        }
        sum += fact;
        n /= 10;
    }
    if (sum == originalNum) {
        printf("Strong number\n");
    } else {
        printf("Not a Strong number\n");
    }
}
```

Output Screenshot:



Section 7: break Statement

41. Print numbers from 1 to 10 but stop when number becomes 5.

Sample Output:

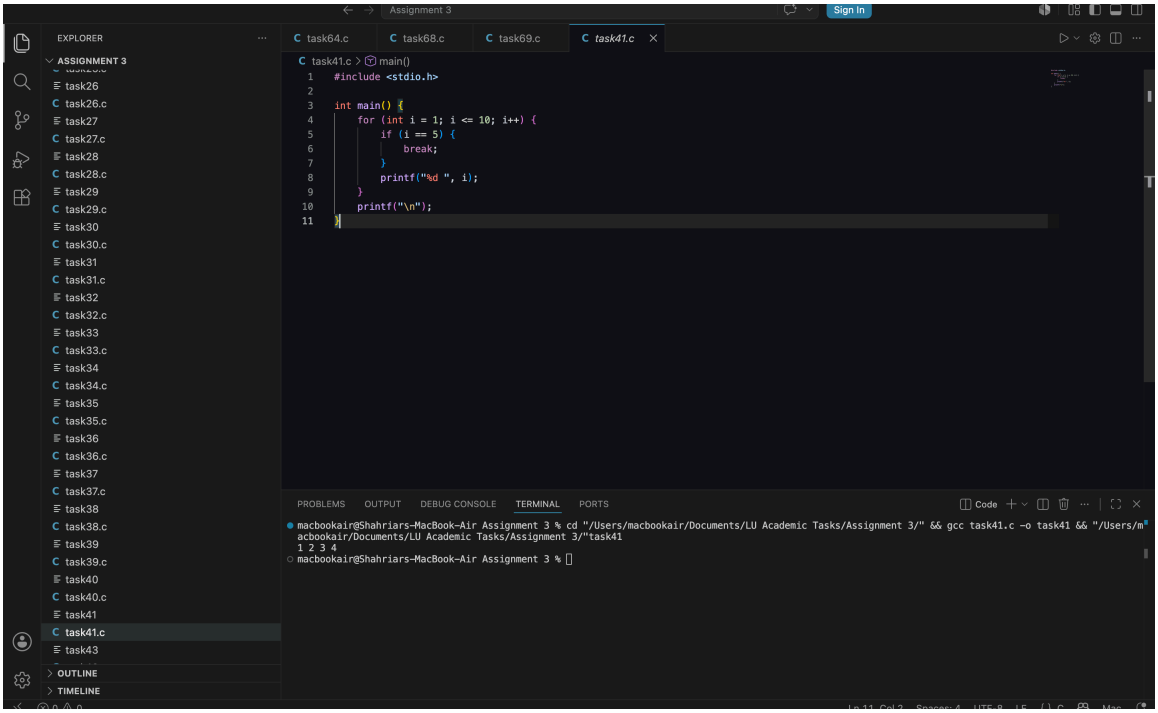
```
1 2 3 4
```

Program:

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 10; i++) {
        if (i == 5) {
            break;
        }
        printf("%d ", i);
    }
    printf("\n");
}
```

Output Screenshot:



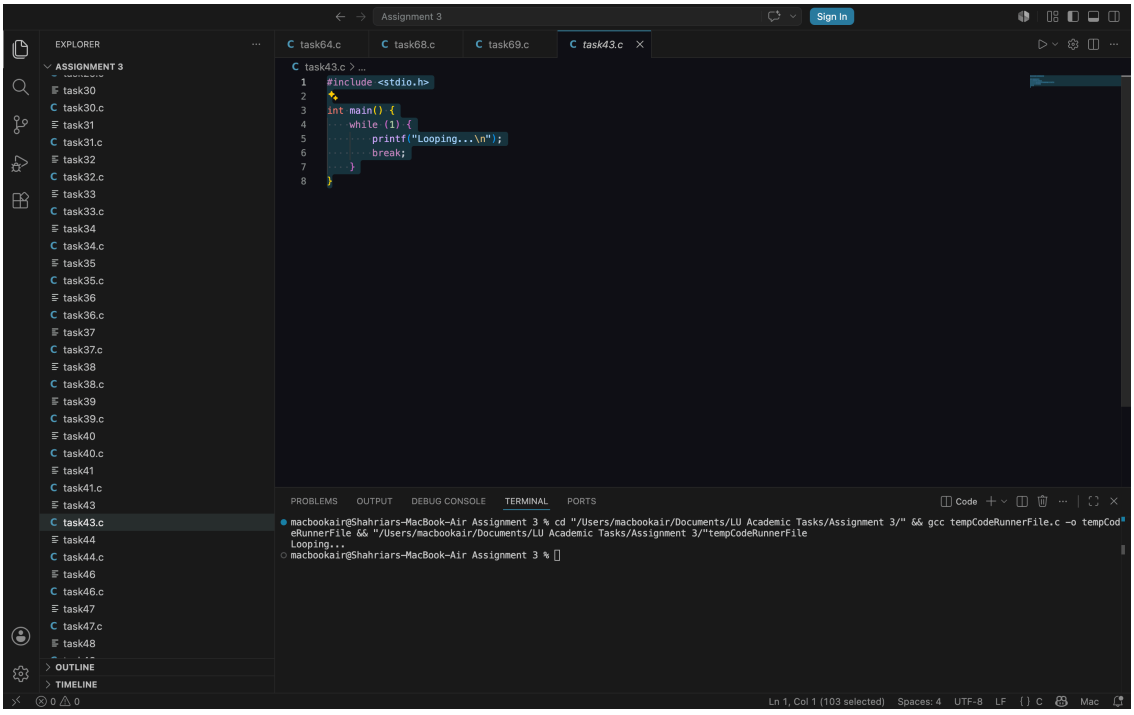
43. Create infinite loop and terminate using `break`.

Program:

```c
#include <stdio.h>

int main() {
    while (1) {
        printf("Looping...\n");
        break;
    }
}
```

Output Screenshot:



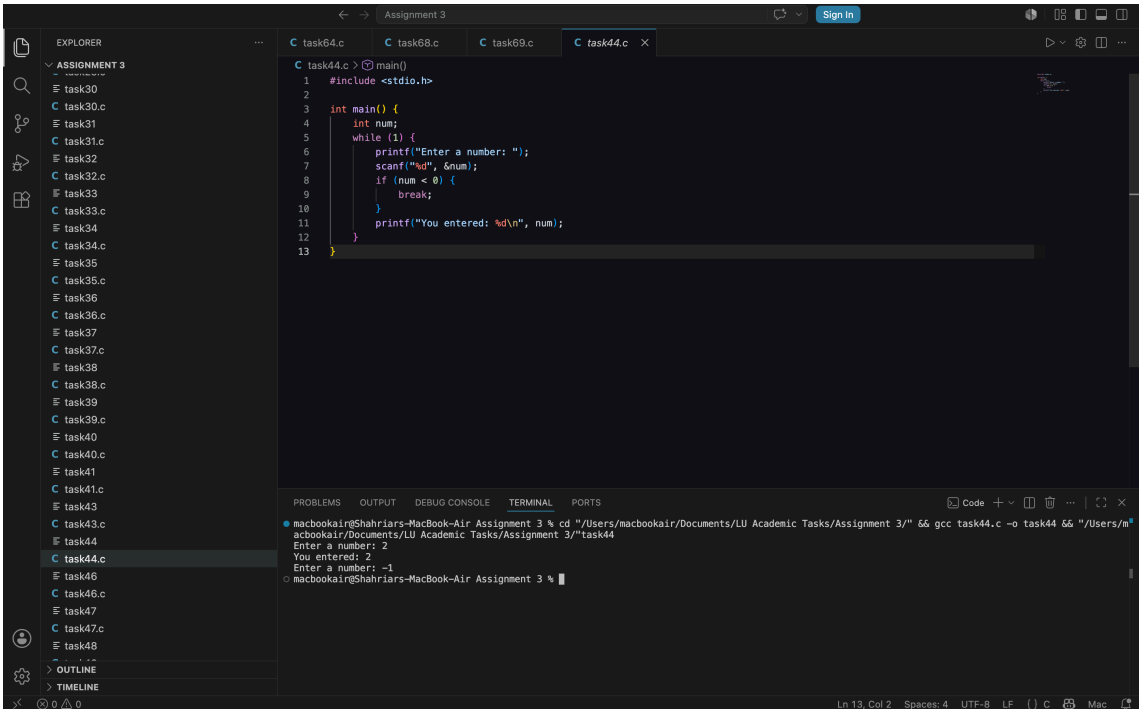
44. Take numbers continuously and stop when negative number is entered.

Program:

```c
#include <stdio.h>

int main() {
    int num;
    while (1) {
        printf("Enter a number: ");
        scanf("%d", &num);
        if (num < 0) {
            break;
        }
        printf("You entered: %d\n", num);
    }
}
```

Output Screenshot:



Section 8: continue Statement

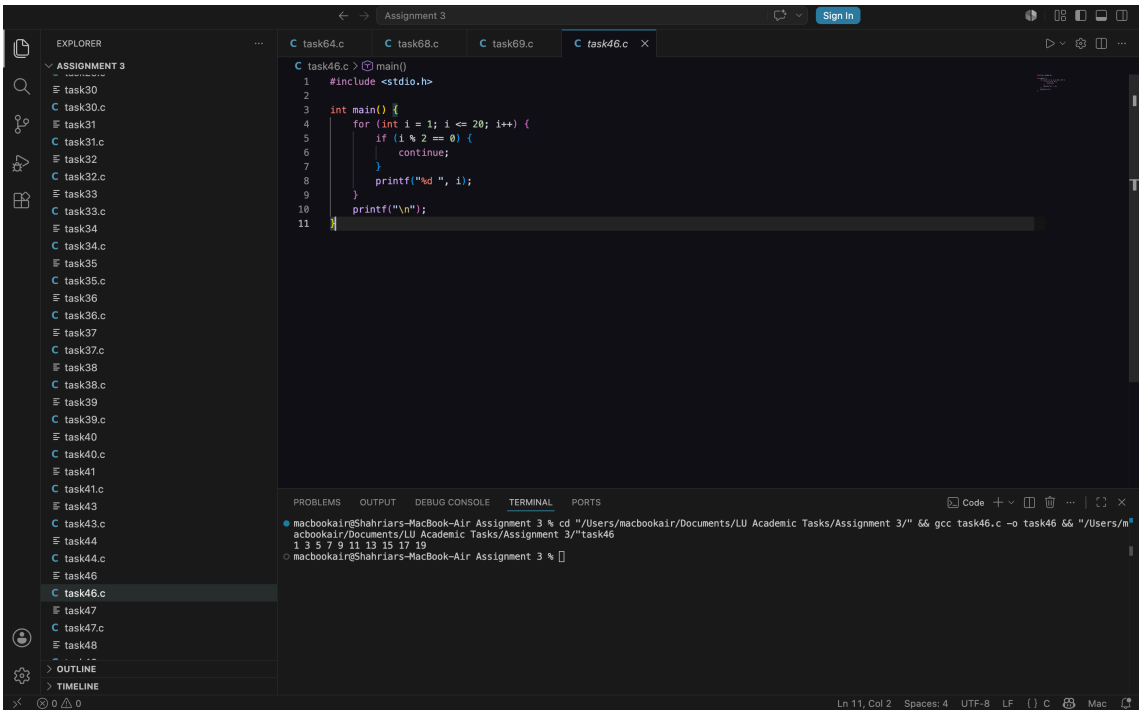
46. Print numbers from 1 to 20 but skip even numbers using `continue`.

Program:

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 20; i++) {
        if (i % 2 == 0) {
            continue;
        }
        printf("%d ", i);
    }
    printf("\n");
}
```

Output Screenshot:



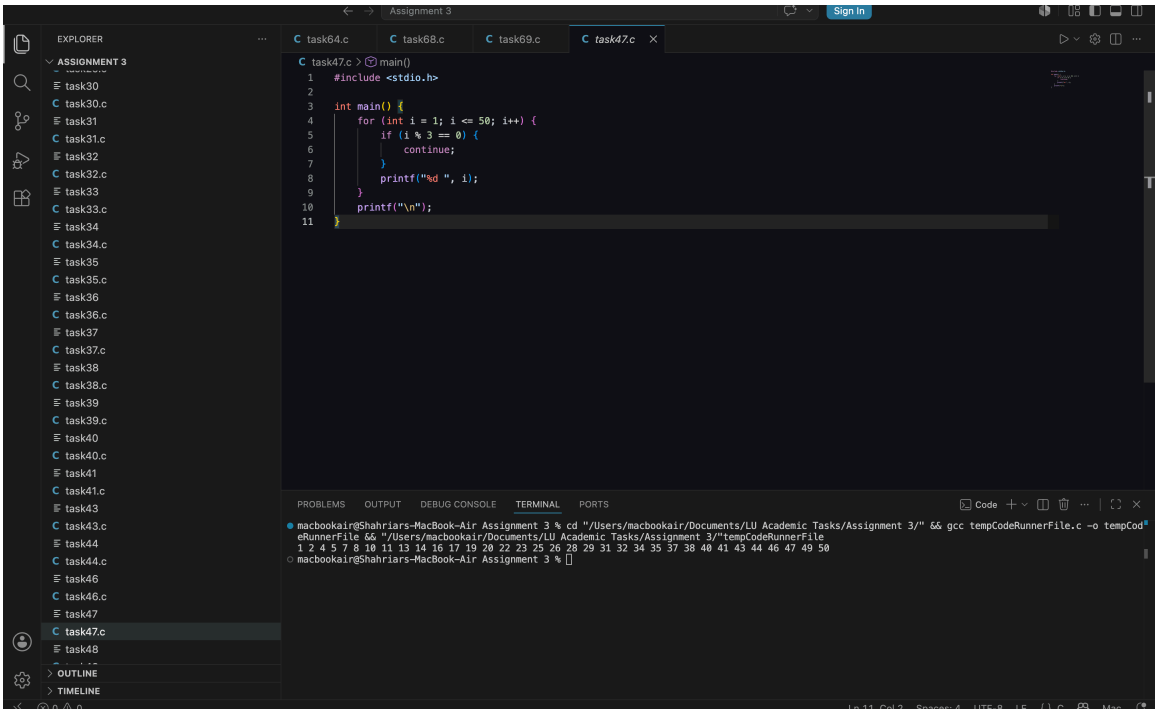
47. Display numbers from 1 to 50 except multiples of 3.

Program:

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 50; i++) {
        if (i % 3 == 0) {
            continue;
        }
        printf("%d ", i);
    }
    printf("\n");
}
```

Output Screenshot:



48. Print all characters except vowels from a string.

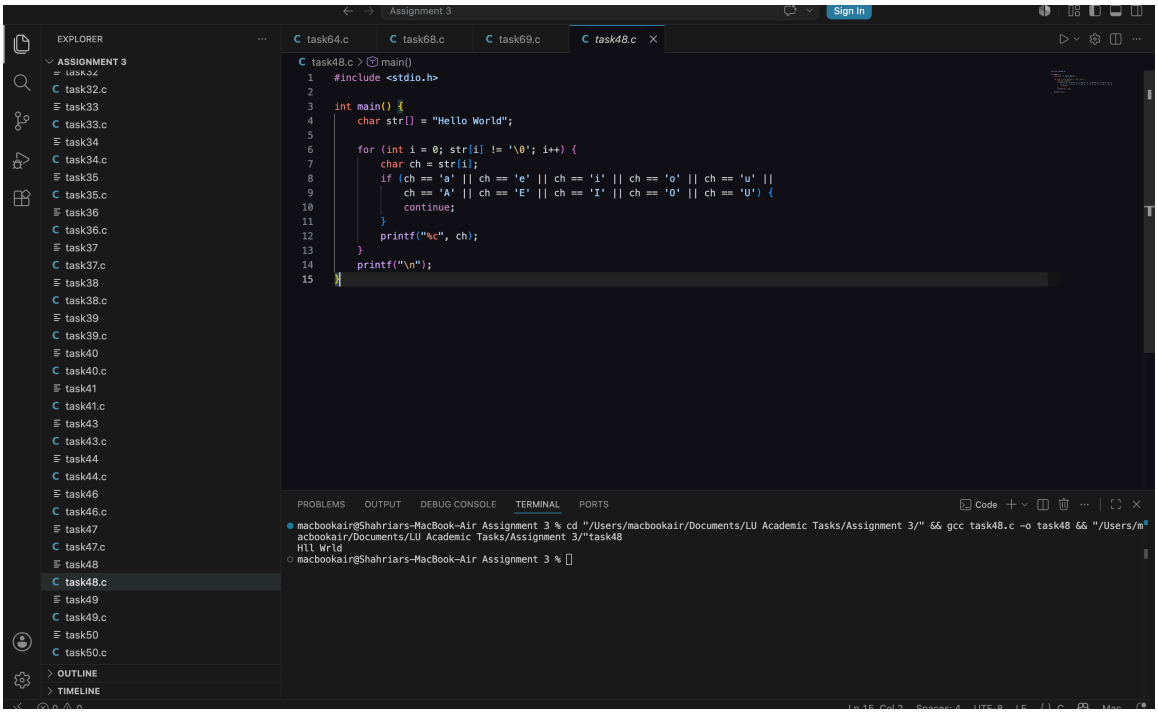
Program:

```c
#include <stdio.h>

int main() {
    char str[] = "Hello World";

    for (int i = 0; str[i] != '\0'; i++) {
        char ch = str[i];
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' ||
ch == 'u' ||
            ch == 'A' || ch == 'E' || ch == 'I' || ch == 'O' ||
ch == 'U') {
            continue;
        }
        printf("%c", ch);
    }
    printf("\n");
}
```

Output Screenshot:



49. Take 10 numbers and print only positive numbers. Skip negative numbers.

Program:

```c
#include <stdio.h>

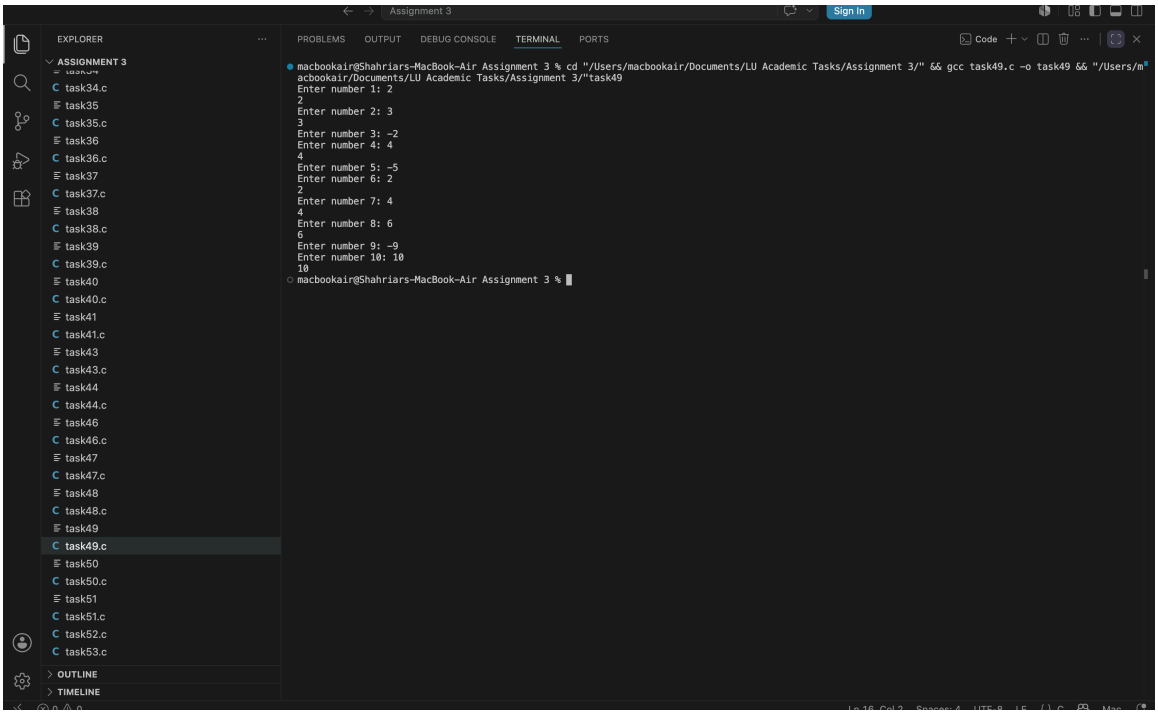
int main() {
    int num;

    for (int i = 1; i <= 10; i++) {
        printf("Enter number %d: ", i);
        scanf("%d", &num);

        if (num < 0) {
            continue;
        }

        printf("%d\n", num);
    }
}
```

Output Screenshot:



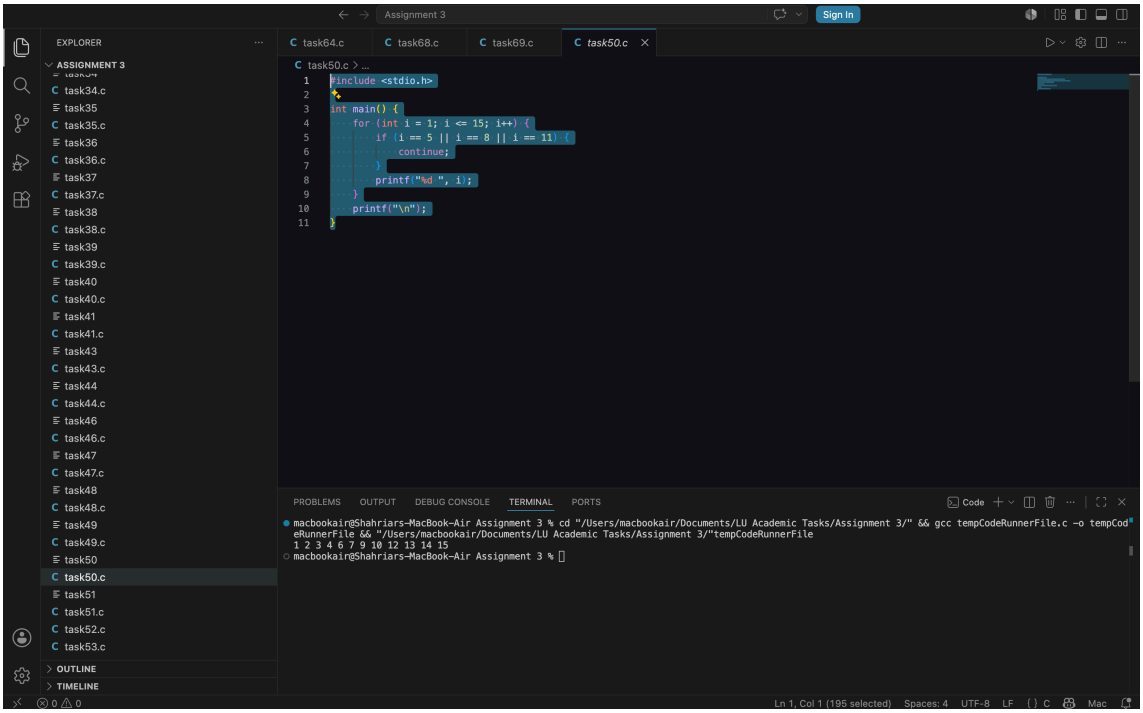
50. Print numbers from 1 to 15 while skipping: `5, 8, 11`

Program:

```c
#include <stdio.h>

int main() {
    for (int i = 1; i <= 15; i++) {
        if (i == 5 || i == 8 || i == 11) {
            continue;
        }
        printf("%d ", i);
    }
    printf("\n");
}
```

Output Screenshot:



Section 9: Infinite Loops & Control Flow

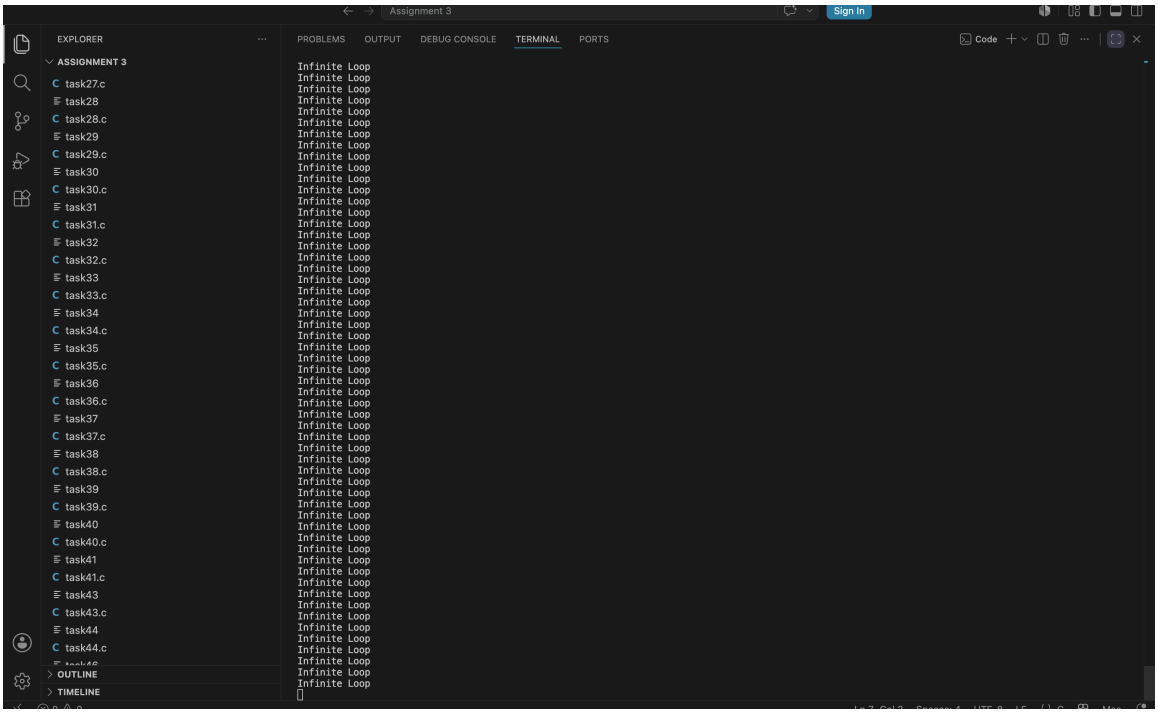
51. Create an infinite loop using: `for(;;)`

Program:

```c
#include <stdio.h>

int main() {
    for (;;) {
        printf("Infinite Loop\n");
    }
}
```

Output Screenshot:



52. Create an infinite loop using: `while(1)`

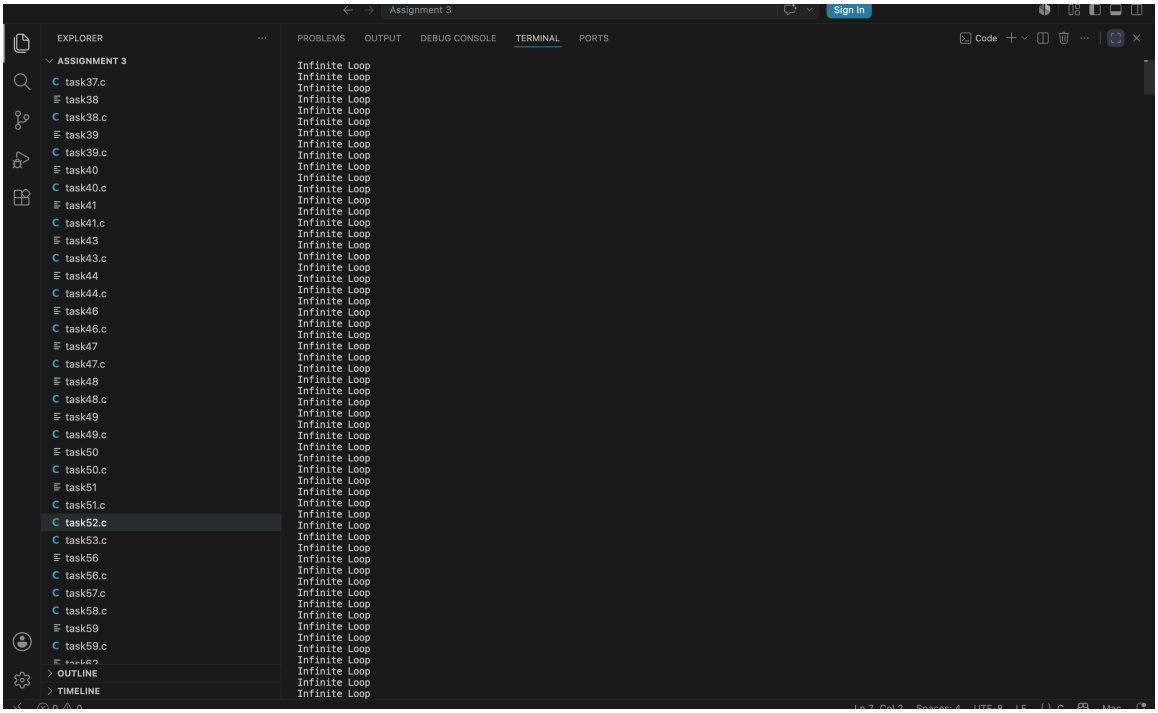
Program:

```c
#include <stdio.h>

int main() {
    while (1) {
```

```
        printf("Infinite Loop\n");
    }
}
```

Output Screenshot:



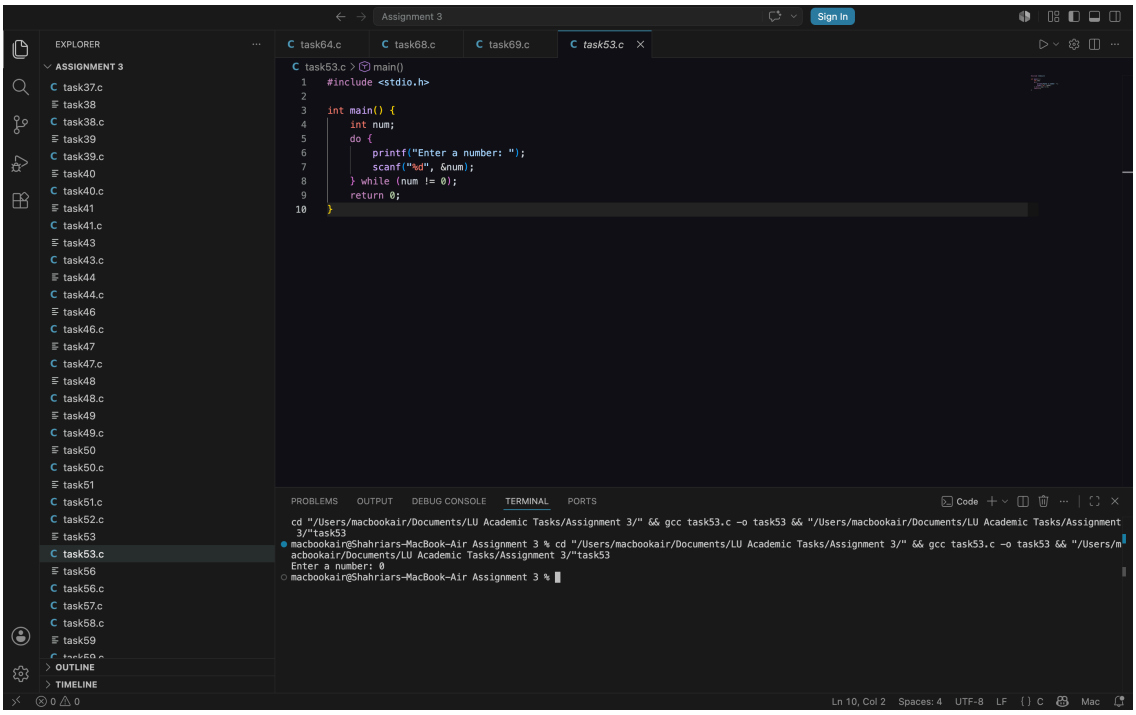
53. Create a loop which exits only when user enters: `0`

Program:

```
#include <stdio.h>

int main() {
    int num;
    do {
        printf("Enter a number: ");
        scanf("%d", &num);
    } while (num != 0);
    return 0;
}
```

Output Screenshot:



Section 10: Mixed Loop Challenges

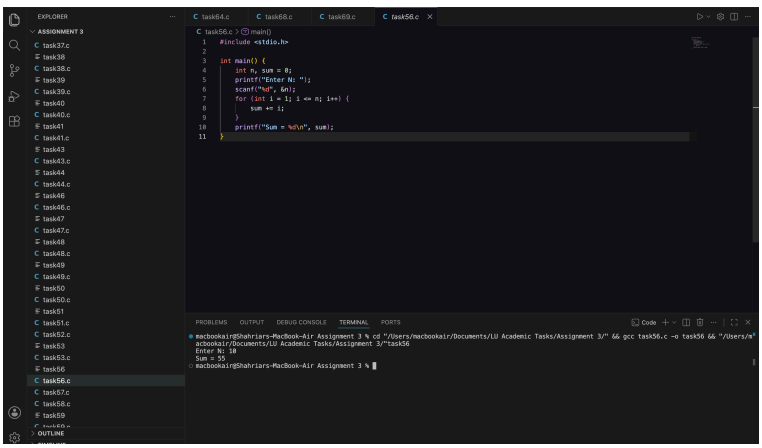
56. Calculate sum of series:
`1 + 2 + 3 + ... + N`

Program:

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter N: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i;
    }
    printf("Sum = %d\n", sum);
}
```

Output Screenshot:



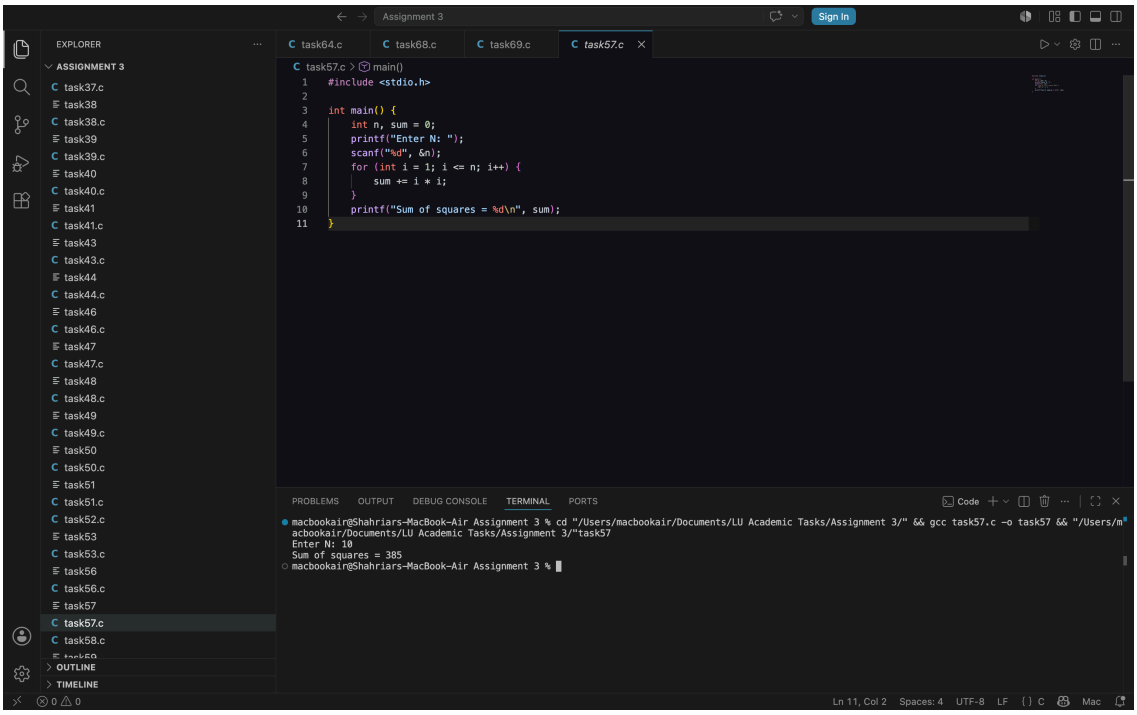
57. Calculate sum of squares series:
$1^2 + 2^2 + 3^2 + ... + N^2$

Program:

```c
#include <stdio.h>

int main() {
    int n, sum = 0;
    printf("Enter N: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += i * i;
    }
    printf("Sum of squares = %d\n", sum);
}
```

Output Screenshot:



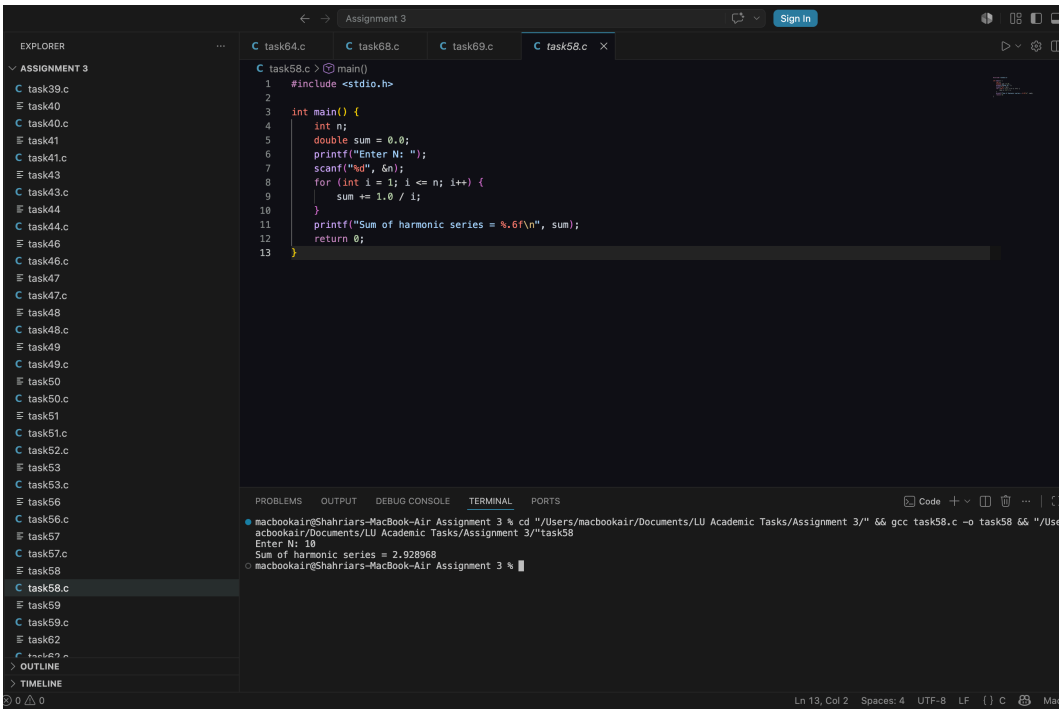
58. Calculate harmonic series:
$1 + 1/2 + 1/3 + ... + 1/N$

Program:

```c
#include <stdio.h>
```

```c
int main() {
    int n;
    double sum = 0.0;
    printf("Enter N: ");
    scanf("%d", &n);
    for (int i = 1; i <= n; i++) {
        sum += 1.0 / i;
    }
    printf("Sum of harmonic series = %.6f\n", sum);
    return 0;
}
```

Output Screenshot:



59. Create combined pattern:

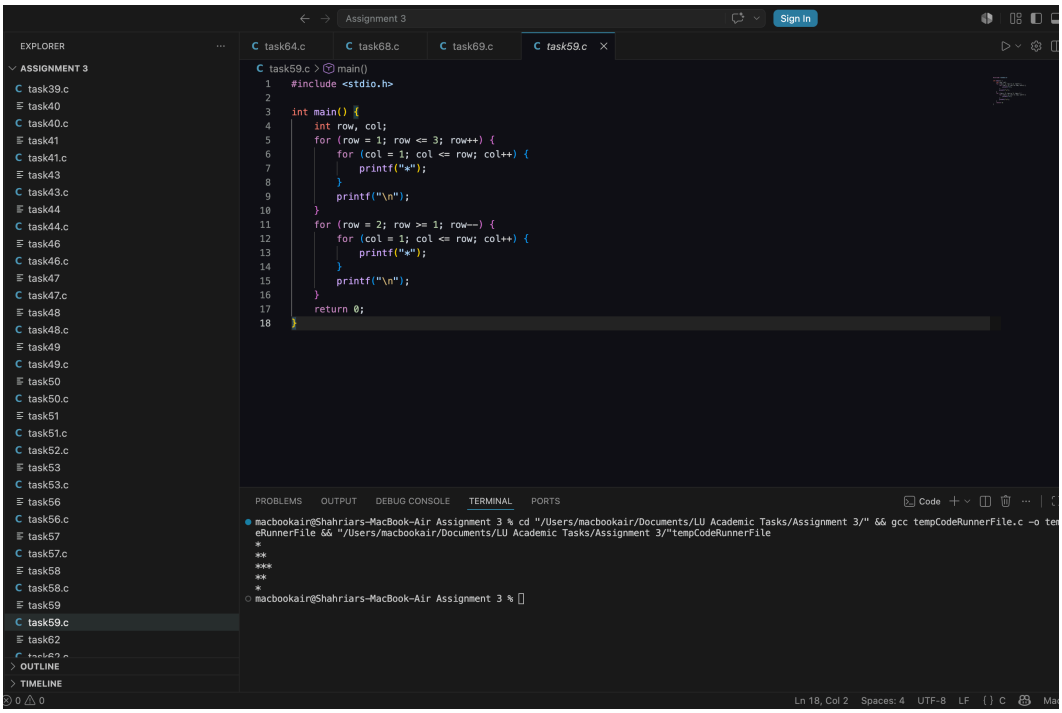
```
*
* *
* * *
* *
*
```

Program:

```c
#include <stdio.h>
```

```c
int main() {
    int row, col;
    for (row = 1; row <= 3; row++) {
        for (col = 1; col <= row; col++) {
            printf("*");
        }
        printf("\n");
    }
    for (row = 2; row >= 1; row--) {
        for (col = 1; col <= row; col++) {
            printf("*");
        }
        printf("\n");
    }
    return 0;
}
```

Output Screenshot:



62. Create Pascal triangle using nested loops.

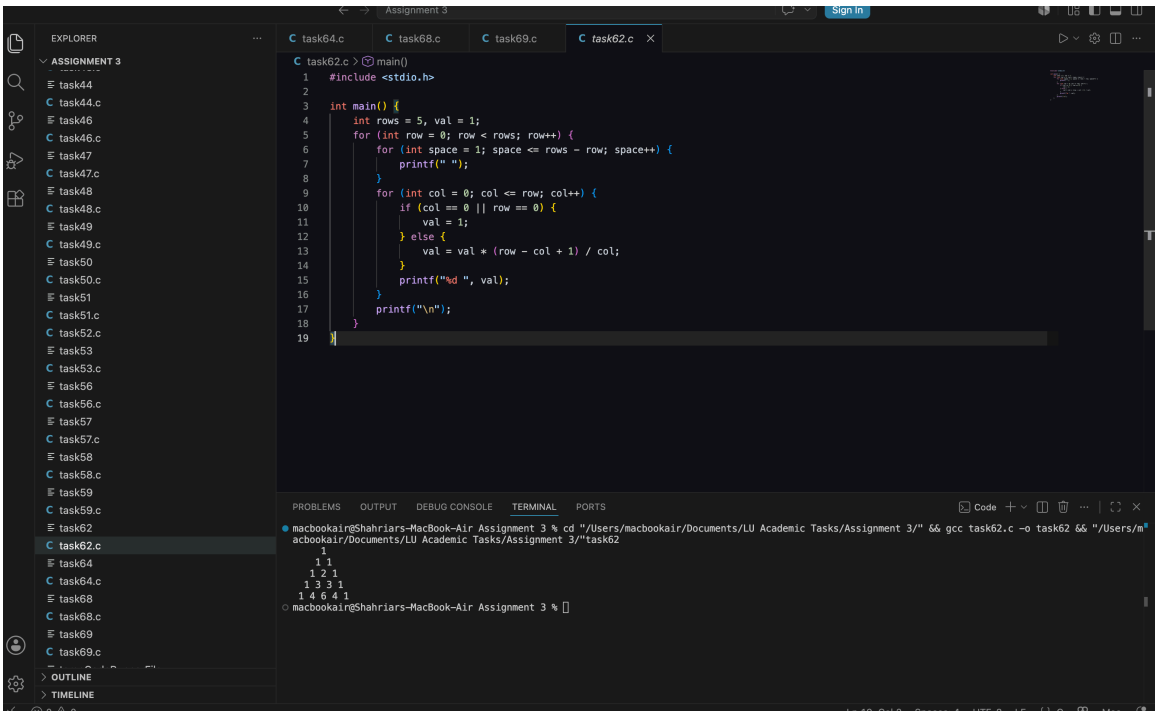
```
    1
   1 1
  1 2 1
 1 3 3 1
1 4 6 4 1
```

Program:

```c
#include <stdio.h>

int main() {
    int rows = 5, val = 1;
    for (int row = 0; row < rows; row++) {
        for (int space = 1; space <= rows - row; space++) {
            printf(" ");
        }
        for (int col = 0; col <= row; col++) {
            if (col == 0 || row == 0) {
                val = 1;
            } else {
                val = val * (row - col + 1) / col;
            }
            printf("%d ", val);
        }
        printf("\n");
    }
}
```

Output Screenshot:



64. Number Pyramid

Print:

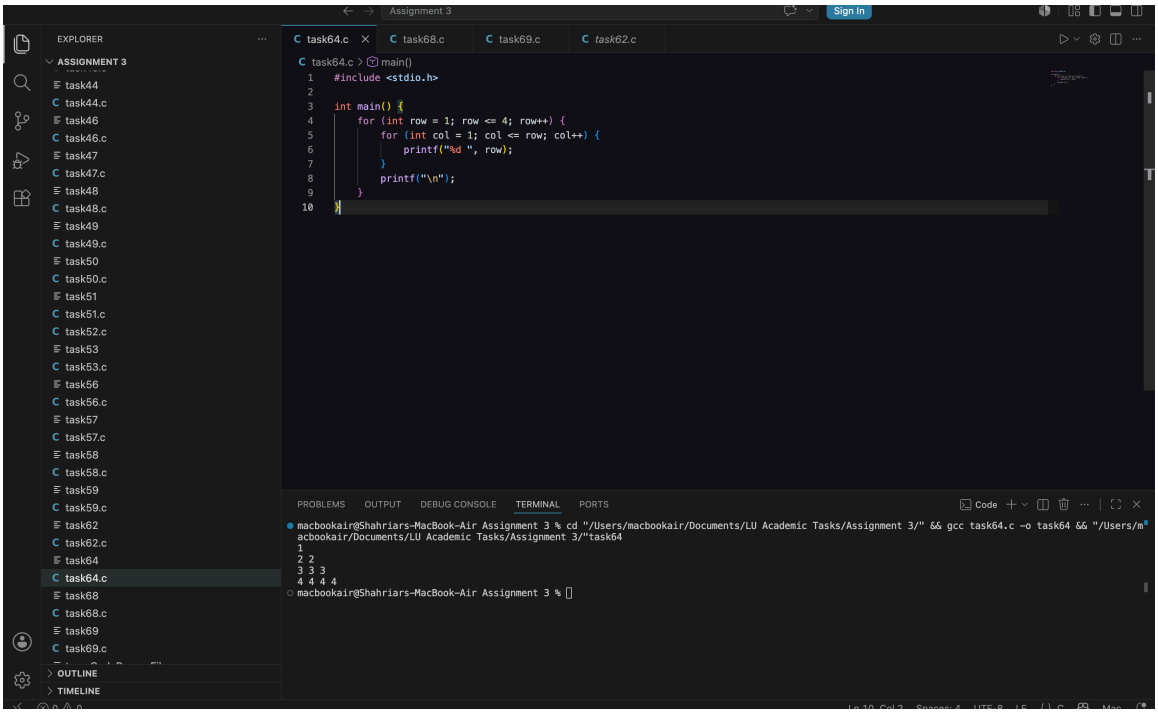
```
1
2 2
3 3 3
4 4 4 4
```

Program:

```c
#include <stdio.h>

int main() {
    for (int row = 1; row <= 4; row++) {
        for (int col = 1; col <= row; col++) {
            printf("%d ", row);
        }
        printf("\n");
    }
}
```

Output Screenshot:



68. Multiplication Triangle

Print:

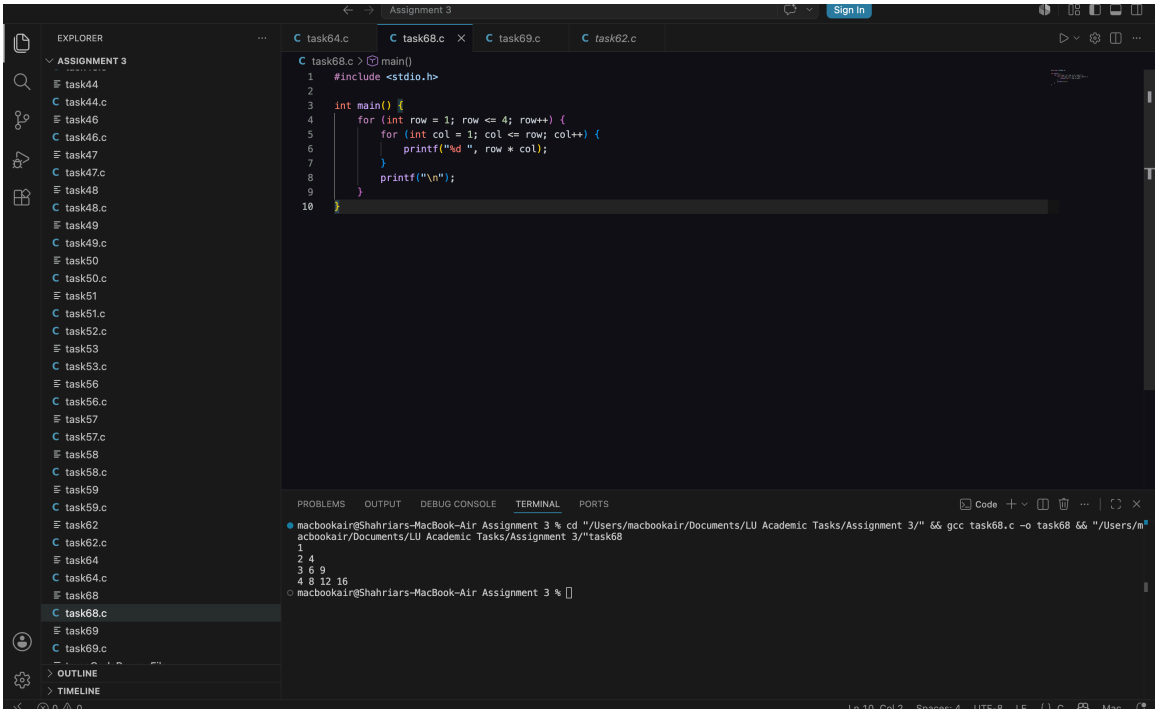
```
1
2 4
3 6 9
4 8 12 16
```

Program:

```c
#include <stdio.h>

int main() {
    for (int row = 1; row <= 4; row++) {
        for (int col = 1; col <= row; col++) {
            printf("%d ", row * col);
        }
        printf("\n");
    }
}
```

Output Screenshot:



69. Print ASCII values from 1 to 127 using loop.

Program:

```
include <stdio.h>

int main() {
    for (int i = 1; i <= 127; i++) {
        printf("ASCII value of %d = %c\n", i, i);
    }
}
```

Output Screenshot:

